

# BG — The classifying stack

## Expository notes

Ignacio Rojas  
ChatGPT (OpenAI)

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*From an ordinary category to a stack, with double covers as a running example*

Developed from Ignacio Rojas's mathematical questions and reading notes,  
with exposition, drafting, and revision by ChatGPT (OpenAI). The  
examples and proofs are intended to make the terminology usable before it  
is compressed into general definitions.

# Contents

|    |                                          |    |
|----|------------------------------------------|----|
| 1  | Introduction and purpose                 | 3  |
| 2  | Starting point: the data defining BG     | 3  |
| 3  | Categories and categories over a base    | 4  |
| 4  | Test schemes and the functor of points   | 13 |
| 5  | Group schemes, built from examples       | 17 |
| 6  | Coverings, sites, and sheaves            | 20 |
| 7  | Torsors: groups without a chosen origin  | 24 |
| 8  | The meaning of the quotient notation     | 27 |
| 9  | Checking that BG is an ordinary category | 29 |
| 10 | The projection to schemes                | 31 |
| 11 | Fiber categories and pullback torsors    | 31 |
| 12 | Cartesian morphisms and the fibration    | 32 |
| 13 | Why the fibers are groupoids             | 34 |
| 14 | The definition of a stack                | 35 |
| 15 | Isomorphisms form a sheaf                | 38 |
| 16 | Effective descent for torsors            | 39 |
| 17 | The stack theorem                        | 42 |
| 18 | A detailed double-cover example          | 43 |
| 19 | Comparison with families of curves       | 44 |
| 20 | Summary of the logical progression       | 47 |
| 21 | Exercises: synthesis and review          | 48 |
| 22 | Hints and solutions                      | 50 |
| 23 | References                               | 57 |

## 1 Introduction and purpose

The aim is to prove that  $BG$  is a stack, starting with its objects and arrows. We first display those data, so that every subsequent definition has a specific purpose: explaining an ingredient of  $(S, P)$  or an operation on such pairs. The preliminary sections are an illustrated guide to reading this definition, not a list of prerequisites to master before meeting  $BG$ .

The reader is assumed to know basic schemes, ring maps, and categories. Group schemes, torsors, sites, and the functor of points are developed here through separate examples and diagrams. In particular, an abstract two-element group will give a family of two labelled copies of each base scheme. We begin the meaning of “two copies” with finite sets.

The exposition follows the categorical viewpoint in Fantechi’s *Stacks for Everybody*, especially Sections 3–4 and 6.3 [2]. Exact external results are cited from the Stacks Project [1]. The final comparison explains how to transfer the argument to families of curves. We prove the stack axioms, not the additional property called algebraicity. Set-theoretic size conventions play no role in the calculations: we use “collection” when the objects need not form a set, just as when speaking of the category of all schemes.

## 2 Starting point: the data defining $BG$

Fix a scheme  $B$  and a group scheme  $G \rightarrow B$ . The example throughout is  $G = (\mathbb{Z}/2\mathbb{Z})_B$ . The following definition names the data first; the subsequent sections unpack *B-scheme*, *group scheme*, *fppf torsor*, and *equivariant*. All torsors in the definition are schemes. The assumptions on  $G$  used in the proof are specified and explained in Section 5.

**Definition 1** (Objects and arrows for  $BG$ ). The data from which we build the total category  $BG$  are as follows.

- ◊ *Objects*: pairs  $(S, P)$ , where  $S \rightarrow B$  is a  $B$ -scheme and  $P \rightarrow S$  is an fppf torsor for  $G_S = G \times_B S$ . The notation  $P$  includes its structure map and its  $G_S$ -action.
- ◊ *Morphisms*: a pair  $(f, \tilde{f})$  from  $(S, P)$  to  $(T, Q)$  consists of a  $B$ -morphism  $f : S \rightarrow T$  and an equivariant scheme morphism  $\tilde{f} : P \rightarrow Q$  making the square commute:

$$\begin{array}{ccc} P & \xrightarrow{\tilde{f}} & Q \\ \downarrow & & \downarrow \\ S & \xrightarrow{f} & T. \end{array}$$

There are three levels in this definition:

- ◊  $B$  is the fixed base *scheme*.
- ◊  $\text{Sch}/B$  is the base *category*; its objects are  $S \rightarrow B$ .

- ◊  $BG$  is the proposed *total category*; its objects carry both  $S$  and a torsor  $P \rightarrow S$ .

Thus  $(S, P)$  is not an object of the base category. The projection will forget  $P$  and retain  $S$ . We have specified objects and arrows; identities, composition, and the category axioms will be checked in Section 9, after the words in the definition have been explained. Defining the data does not yet establish those axioms.

### 2.1 Roadmap for the proof

Here is the route we will follow to show that  $BG$  is a stack.

- (1)  $BG$  is a category: define identities and composition and check their laws.
- (2)  $BG$  is a category over  $\text{Sch}/B$ : verify the projection functor.
- (3)  $BG$  is fibered over  $\text{Sch}/B$ : construct pullbacks and prove their cartesian universal property.
- (4)  $BG$  is fibered in groupoids: prove that every arrow over an identity in the base is invertible.
- (5) Isomorphisms between torsors form a sheaf: glue compatible local isomorphisms uniquely.
- (6) Object descent is effective: construct a global torsor from compatible local torsors.

- (7) Collect these results to conclude that  $BG$  is a stack.

Each numbered item needs its own verification; the last two gluing steps are additional to constructing a fibered category.

### 2.2 Guide to the ingredients

- ◊ Section 3:  $B$ -schemes, categories over a base, disjoint unions, and fiber products.
- ◊ Section 4: how to interpret points of a scheme as maps, with affine coordinates and a proof of the Yoneda argument used.
- ◊ Sections 5–7: the group scheme  $G$ , the covering rule, and the torsor  $P \rightarrow S$ .
- ◊ Section 8: why the notation  $[\text{pt}/G]$  leads to these objects.

After this introduction to the ingredients, Section 9 returns to Definition 1 and starts the proof in the order above. Exercises 25–30 provide additional practice with the preliminary constructions.

## 3 Categories over a base

### 3.1 Schemes and schemes with a specified base

A scheme  $S$  has an underlying topological space and a structure sheaf, locally isomorphic as a locally ringed space to  $\text{Spec } A$  for a ring  $A$ . The topology and sheaf are not arbitrary extra choices. A

$B$ -scheme is this same kind of scheme together with a specified morphism  $s : S \rightarrow B$ . The added structure is the map, not a new meaning of the word scheme.

**Example 1** (The affine line as a scheme over a field). Let  $k$  be a field and take  $B = \operatorname{Spec} k$ . The inclusion of constants  $\iota : k \rightarrow k[t]$  induces a scheme morphism in the reverse direction:

$$k \xrightarrow{\iota} k[t] \quad c \mapsto c,$$

$$\mathbb{A}_k^1 = \operatorname{Spec} k[t] \xrightarrow{\pi = \operatorname{Spec} \iota} \operatorname{Spec} k.$$

What does  $\pi$  do? A topological point of  $\operatorname{Spec} k[t]$  is a prime ideal  $\mathfrak{p}$ . The underlying map sends it to  $\iota^{-1}(\mathfrak{p}) = \mathfrak{p} \cap k$ . This is always  $(0)$ : a proper ideal cannot contain a nonzero constant, which is a unit. Since  $(0)$  is the only prime ideal of a field, every point maps to the unique point of  $\operatorname{Spec} k$ . For instance,  $(t - a)$  maps to  $(0)$ .

The scheme morphism also includes a map on functions: a constant  $c$  on the base pulls back to the constant polynomial  $c$ . Thus the map of spectra and the inclusion of constants describe the same morphism from its two sides. The general scheme–ring correspondence is recalled in Section 4.

**Example 2** (The relative affine line and its local squares). For a general base scheme  $B$ , the relative affine line comes with a map  $\pi : \mathbb{A}_B^1 \rightarrow B$ . Over an affine open

$U = \operatorname{Spec} A \subset B$ , its description is

$$\begin{array}{ccc} \operatorname{Spec} A[t] & \hookrightarrow & \mathbb{A}_B^1 \\ \pi_U \downarrow & & \downarrow \pi \\ U = \operatorname{Spec} A & \hookrightarrow & B. \end{array}$$

The left vertical map is induced by  $A \rightarrow A[t]$ ,  $a \mapsto a$ ; on prime ideals it sends  $\mathfrak{p}$  to  $\mathfrak{p} \cap A$ . Unlike the field example, the target can have many points. The square says that  $\operatorname{Spec} A[t]$  is the inverse image of  $U$  under  $\pi$ . It is a fiber-product square, in the sense explained below.

For a principal open  $D(a) \subset U$ , the smaller local square is

$$\begin{array}{ccc} \operatorname{Spec} A_a[t] & \hookrightarrow & \operatorname{Spec} A[t] \\ \downarrow & & \downarrow \\ \operatorname{Spec} A_a & \hookrightarrow & \operatorname{Spec} A. \end{array}$$

The horizontal maps come from localization, and the vertical maps from inclusion of constants. The identity  $A[t]_a = A_a[t]$  makes these descriptions agree on smaller opens. They glue to the relative affine line. This is the affine-local description of Definition 27.5.1, Tag 01M0, with the restriction and base-change properties in Lemma 27.4.6, Tag 01LX.

**Definition 2** (The category of  $B$ -schemes). The category  $\operatorname{Sch}/B$  is defined by the following data.

- ◇ *Objects*: pairs  $(S, s)$  with  $S$  a scheme and  $s : S \rightarrow B$  a scheme morphism.

- ◇ *Morphisms*: an arrow  $(S, s) \rightarrow (T, t)$  is a scheme morphism  $f : S \rightarrow T$  such that  $t \circ f = s$ .
- ◇ *Identities and composition*: use the identity and composition of scheme morphisms. The equation defining an arrow is preserved: if  $u \circ g = t$  and  $t \circ f = s$ , then  $u \circ (g \circ f) = s$ .

The condition on morphisms is the commuting triangle

$$\begin{array}{ccc} S & \xrightarrow{f} & T \\ & \searrow s & \swarrow t \\ & B & \end{array}$$

The identity and associativity laws are inherited from Sch.

*Remark 1* (Comparison with Fantechi). This is exactly the slice construction in Fantechi's Section 3.2, with  $B$  in place of her fixed scheme  $S$ ; it is also called the *slice category*. We shall often abbreviate its object  $(S, s)$  to  $S$ , but the map  $s$  remains part of the data. "A scheme over  $B$ " and "a  $B$ -scheme" mean the same thing. The object  $(B, \text{id}_B)$  is terminal: from  $(S, s)$  there is exactly one arrow to it, namely  $s$ .

### 3.2 Categories over a category, and their fibers

**Definition 3.** A category consists of objects, sets of arrows between objects, identity arrows, and an associative composition law with left and right identity laws.

A functor preserves the sources and targets of arrows, identities, and composition.

**Definition 4.** A *category over a category*  $\mathcal{C}$  is a category  $\mathcal{F}$  equipped with a functor  $p : \mathcal{F} \rightarrow \mathcal{C}$ . An object  $x$  lies over  $S$  if  $p(x) = S$ ; an arrow  $u : x \rightarrow y$  lies over  $f : S \rightarrow T$  if  $p(u) = f$ . The *fiber category*  $\mathcal{F}_S$  is specified by

$$\text{Ob}(\mathcal{F}_S) = \{x : p(x) = S\}$$

and, for two such objects  $x, y$ , by

$$\text{Hom}_{\mathcal{F}_S}(x, y) = \{u \in \text{Hom}_{\mathcal{F}}(x, y) : p(u) = \text{id}_S\}.$$

These arrows are called *vertical*.

We use  $\mathcal{F}_S$  for the fiber throughout, and  $(BG)_S$  for the fiber of  $BG$ . One may also write  $p^{-1}(S)$  provided that its arrows mean exactly those in the definition, not all arrows between the selected objects. Later, parentheses in  $X(T)$  will denote a *set of points*, not a fiber category.

*Remark 2* (Vertical arrows). "Vertical" is a geometric nickname: imagine objects of  $\mathcal{F}$  arranged above the objects of  $\mathcal{C}$ . A vertical arrow does not move in the base; its base arrow is the identity. It need not be drawn vertically on the page. For a general functor there is no additional triangle to check:  $p(u) = \text{id}_S$  is the complete condition. Triangles arise when the particular categories have themselves been defined using triangles, as with  $\text{Sch}/B$ .

**Exercise 1** (Fiber versus full subcategory). A *full subcategory* on selected objects retains every arrow of the original category between those objects. Let  $\mathcal{D}$  be the full subcategory of  $\mathcal{F}$  on the objects  $x$  with  $p(x) = S$ . Write the object collections and arrow sets of  $\mathcal{D}$  and  $\mathcal{F}_S$  separately. Then take  $p = \text{id}_c$  and an endomorphism  $\varepsilon : S \rightarrow S$  with  $\varepsilon \neq \text{id}_S$ . Which category contains  $\varepsilon$ ? Why? Use  $t \mapsto t^2$  on  $\mathbb{A}_k^1$  as a concrete example. The full comparison is worked out in the solution on page 50.

### 3.3 Fiber products: construction, then universal property

Suppose given  $q : X \rightarrow S$  and  $f : T \rightarrow S$ . We want a scheme over  $T$  that retains precisely the part of  $X$  lying over the base specified by  $f$ . This operation is called *base change* or *pullback*.

**Example 3** (A fiber product of finite sets). Take three sets

$$S = \{0, 1\}, \quad T = \{a, b, c\}, \\ X = \{u, v, w\}.$$

Define maps by

$$\begin{array}{c|ccc} t & a & b & c \\ \hline f(t) & 0 & 0 & 1 \end{array} \quad \begin{array}{c|ccc} x & u & v & w \\ \hline q(x) & 0 & 1 & 1. \end{array}$$

The fiber product retains only pairs with the same image in  $S$ :

$$T \times_S X = \{(t, x) : f(t) = q(x)\} \\ = \{(a, u), (b, u), (c, v), (c, w)\}.$$

For example,  $(a, v)$  is excluded because  $0 \neq 1$ . The projections remember the first and second entries:

$$\begin{array}{ccc} T \times_S X & \xrightarrow{\text{pr}_X} & X \\ \text{pr}_T \downarrow & & \downarrow q \\ T & \xrightarrow{f} & S. \end{array}$$

Over  $a$  there is one choice,  $u$ ; over  $c$  there are two choices,  $v$  and  $w$ . Pullback replaces the old base labels  $0, 1$  by the new base labels  $a, b, c$ , using  $f$  to decide which old fiber to copy.

**Example 4** (A fiber product of affine schemes). For schemes, structure sheaves must be retained as well. If  $S = \text{Spec } R$ ,  $T = \text{Spec } A$ , and  $X = \text{Spec } C$ , use the two ring maps  $\alpha : R \rightarrow A$  and  $\gamma : R \rightarrow C$ . Form

$$T \times_S X = \text{Spec}(A \otimes_R C).$$

The ring diagram is

$$\begin{array}{ccc} R & \xrightarrow{\gamma} & C \\ \alpha \downarrow & & \downarrow c \mapsto 1 \otimes c \\ A & \xrightarrow{a \mapsto a \otimes 1} & A \otimes_R C. \end{array}$$

Applying  $\text{Spec}$  reverses all four arrows:

$$\begin{array}{ccc} \text{Spec}(A \otimes_R C) & \xrightarrow{\text{pr}_X} & \text{Spec } C \\ \text{pr}_T \downarrow & & \downarrow q \\ \text{Spec } A & \xrightarrow{f} & \text{Spec } R. \end{array}$$

The ring square commutes because  $\alpha(r) \otimes 1 = 1 \otimes \gamma(r)$ . The scheme square therefore commutes too. For example, if  $R = k$ ,

$A = k[v]$ , and  $C = k[x]$ , then  $A \otimes_R C \cong k[v, x]$ : the product of two affine lines over  $k$  is the affine plane, with its coordinate projections. Lemma 26.17.2, Tag 01JQ, proves the general affine formula.

These affine constructions glue to give fiber products of arbitrary schemes (Section 26.17, Tag 01J0). If the original arrows are  $B$ -morphisms, the product is a  $B$ -scheme via its map to  $S \rightarrow B$ .

**Definition 5** (The universal property of a fiber product). The universal property describes exactly the maps *into this constructed scheme*. For every scheme  $W$  with maps  $b : W \rightarrow T$  and  $a : W \rightarrow X$  satisfying  $fb = qa$ , there exists exactly one map  $\lambda : W \rightarrow T \times_S X$  with

$$\mathrm{pr}_T \lambda = b, \quad \mathrm{pr}_X \lambda = a.$$

The diagram is

$$\begin{array}{ccccc} W & & & & \\ & \searrow \lambda & & \searrow a & \\ & T \times_S X & \xrightarrow{\mathrm{pr}_X} & X & \\ & \downarrow \mathrm{pr}_T & & \downarrow q & \\ & T & \xrightarrow{f} & S & \end{array}$$

Thus a compatible pair of maps from  $W$  gives a *unique map*, not generally an isomorphism.

**Example 5** (The unique map need not be an isomorphism). Let the common target

be  $S = \mathrm{Spec} k$ , and take  $T = \mathrm{Spec} k[v]$  and  $X = \mathrm{Spec} k[x]$ . Put  $W = \mathrm{Spec} k$ . Define  $b : W \rightarrow T$  by  $v \mapsto 0$  on rings and  $a : W \rightarrow X$  by  $x \mapsto 0$ . Both composites  $W \rightarrow S$  are the identity. The resulting diagram is

$$\begin{array}{ccc} \mathrm{Spec} k & & \\ & \searrow \lambda & \searrow a \\ & \mathbb{A}_k^2 & \xrightarrow{\mathrm{pr}_X} \mathbb{A}_{x,k}^1 \\ & \downarrow \mathrm{pr}_T & \downarrow \\ & \mathbb{A}_{v,k}^1 & \rightarrow \mathrm{Spec} k. \end{array}$$

The map  $\lambda$  is the origin in the plane: its ring map is  $k[v, x] \rightarrow k$ ,  $F \mapsto F(0, 0)$ . There is exactly one map with these two coordinate maps, because the images of  $v$  and  $x$  determine the ring homomorphism. It is not an isomorphism: that homomorphism has the nonzero kernel  $(v, x)$ .

The point  $W$  does not itself have the universal property. For instance, the two maps from  $Z = \mathrm{Spec} k[z]$  given by  $v \mapsto z$  and  $x \mapsto 0$  are compatible over  $k$ , but cannot factor through  $W$  with its origin projections. Such a factorization would force the image of  $v$  to be 0, not  $z$ .

**Proposition 1** (Uniqueness of a fiber product). Suppose  $F$ , with projections  $p_T, p_X$ , and  $W'$ , with projections  $b', a'$ , both satisfy the fiber-product universal property for  $T \xrightarrow{f} S \xleftarrow{q} X$ . There is a unique isomorphism  $\alpha : W' \rightarrow F$  preserving both projections.

*Proof.* First use  $W'$  as the source in the universal property of  $F$ . It gives the dashed map  $\alpha$  in

$$\begin{array}{ccc}
 W' & \xrightarrow{a'} & X \\
 \downarrow b' & \searrow \alpha & \nearrow p_X \\
 & F & \\
 \nearrow p_T & & \downarrow q \\
 T & \xrightarrow{f} & S
 \end{array}$$

Thus  $p_T \alpha = b'$  and  $p_X \alpha = a'$ . Now use  $F$  as the source in the universal property of  $W'$ :

$$\begin{array}{ccc}
 F & \xrightarrow{p_X} & X \\
 \downarrow p_T & \searrow \beta & \nearrow a' \\
 & W' & \\
 \nearrow b' & & \downarrow q \\
 T & \xrightarrow{f} & S
 \end{array}$$

This gives  $b' \beta = p_T$  and  $a' \beta = p_X$ . Consequently

$$p_T \alpha \beta = p_T, \quad p_X \alpha \beta = p_X.$$

Both  $\alpha \beta$  and  $\text{id}_F$  are maps  $F \rightarrow F$  with these projections. Uniqueness in the universal property of  $F$  gives  $\alpha \beta = \text{id}_F$ . The same argument with  $W'$  gives  $\beta \alpha = \text{id}_{W'}$ . Thus  $\alpha$  and  $\beta$  are inverse. The first universal property already says that  $\alpha$  is the only map preserving the projections, hence the only such isomorphism.  $\square$

*Remark 3.* Merely having the outer commuting square for  $W'$  gives  $\alpha$ , but not  $\beta$ . The second universal property, valid for *every* source scheme, is what supplies the inverse.

**Example 6** (Pulling back a parabola). Let  $k$  be a field. Name the original base and the family:

$$\begin{aligned}
 S &= \text{Spec } k[t], \\
 X &= \text{Spec}(k[t, x]/(x^2 - t)).
 \end{aligned}$$

Thus  $X$  is the closed subscheme  $V(x^2 - t)$  of the affine  $(t, x)$ -plane. The map  $q : X \rightarrow S$  is induced by  $k[t] \rightarrow k[t, x]/(x^2 - t)$ ,  $t \mapsto \bar{t}$ . On  $k$ -valued points it is precisely  $(t, x) \mapsto t$ . Over a value  $t_0$ , its fiber has equation  $x^2 = t_0$ . For  $k = \mathbb{R}$  the picture is the parabola  $t = x^2$  projecting to the  $t$ -axis.

Now choose a *new base*  $T = \text{Spec } k[v]$  and a map  $f : T \rightarrow S$  whose ring homomorphism is

$$\alpha : k[t] \longrightarrow k[v], \quad t \longmapsto v^2.$$

On  $k$ -valued points,  $f$  sends  $v$  to  $v^2$ . The pullback must assign to  $v$  the old fiber over  $t = v^2$ : its equation is therefore  $x^2 = v^2$ . Here is exactly how the earlier tensor product produces that equation. In Example 4, take

$$\begin{aligned}
 R &= k[t], & A &= k[v], \\
 C &= k[t, x]/(x^2 - t).
 \end{aligned}$$

Use  $\alpha$  above for  $R \rightarrow A$ , and  $t \mapsto \bar{t}$  for  $R \rightarrow C$ . Then

$$\begin{aligned}
 A \otimes_R C &\cong k[v, t, x]/(t - v^2, x^2 - t) \\
 &\cong k[v, x]/(x^2 - v^2).
 \end{aligned}$$

The two relations first identify the copies of  $t$ , then retain the equation defining  $X$ . Eliminating  $t$  gives the second isomorphism. Write  $X' = \operatorname{Spec} k[v, x]/(x^2 - v^2)$ . The scheme square is

$$\begin{array}{ccc} X' & \xrightarrow{h} & X \\ q' \downarrow & & \downarrow q \\ T & \xrightarrow{f} & S. \end{array}$$

The maps, on  $k$ -valued points, are

$$\begin{aligned} h(v, x) &= (v^2, x), & q'(v, x) &= v, \\ q(t, x) &= t, & f(v) &= v^2. \end{aligned}$$

In particular,  $qh(v, x) = v^2 = fq'(v, x)$ . On coordinate rings,  $h$  sends  $\bar{t}$  to  $v^2$  and  $\bar{x}$  to  $x$ .

The real picture suggested by the equations is accurate: the original family is a parabola, while the pulled-back family is the union of the lines  $x = v$  and  $x = -v$ . But the operation is not replacing one curve by an isomorphic curve. It changes the base map and retains a comparison map  $h$  to the original family. For instance, over  $v = 2$  the new fiber copies the old fiber over  $t = 4$ , with choices  $x = 2, -2$ .

If  $\operatorname{char} k \neq 2$ , restrict to  $v \neq 0$ . In  $k[v, v^{-1}, x]$ , the ideals  $(x - v)$  and  $(x + v)$  are comaximal: their difference contains the unit  $2v$ . The Chinese remainder theorem therefore gives

$$\frac{k[v, v^{-1}, x]}{(x^2 - v^2)} \cong k[v, v^{-1}] \times k[v, v^{-1}].$$

This is where the two lines become two *disjoint* copies of the new base. Over  $v = 0$

they meet, so the full family is not the trivial double cover. In characteristic 2, the equation is  $(x - v)^2 = 0$ ; there are not two distinct lines. These qualifications matter when we later compare double covers with torsors.

### 3.4 Two copies are a disjoint union, not a product

**Example 7** (Two labelled copies of a three-element set). Let  $E = \{1, 2, 3\}$ . The disjoint union  $E \amalg E$  is

$$\begin{aligned} E \amalg E &= (E \times \{0\}) \cup (E \times \{1\}) \\ &= \{1_0, 2_0, 3_0, 1_1, 2_1, 3_1\}, \end{aligned}$$

where  $n_i$  abbreviates  $(n, i)$ . In particular  $1_0 \neq 1_1$ : the label is part of the element. This is different from  $E \cup E = E$ , which has three elements, and from  $E \times E$ , which has nine.

To give a function  $F : E \amalg E \rightarrow E$ , specify its value on each of the six labelled elements. Equivalently, give two functions  $f_0, f_1 : E \rightarrow E$  and set

$$F(n_0) = f_0(n), \quad F(n_1) = f_1(n).$$

For example, choose  $f_0 = \operatorname{id}_E$  and  $f_1(n) = 4 - n$ . The function is

|        |       |       |       |       |       |       |
|--------|-------|-------|-------|-------|-------|-------|
| $y$    | $1_0$ | $2_0$ | $3_0$ | $1_1$ | $2_1$ | $3_1$ |
| $F(y)$ | 1     | 2     | 3     | 3     | 2     | 1     |

The particular choice  $f_0 = f_1 = \operatorname{id}_E$  gives the *fold map*  $\pi(n_i) = n$ . It forgets only the label. Both  $2_0$  and  $2_1$ , for instance, map to 2.

**Definition 6** (Disjoint union and its structure sheaf). For a scheme  $S$ , construct  $S \sqcup S$  with underlying set  $|S| \times \{0, 1\}$ . Write its points as  $s_i = (s, i)$ , with  $s \in |S|$ . An open subset is of the form  $U_0 \sqcup U_1$ , where each  $U_i$  is open in  $S$ . Define

$$\mathcal{O}_{S \sqcup S}(U_0 \sqcup U_1) = \mathcal{O}_S(U_0) \times \mathcal{O}_S(U_1),$$

with restriction on each factor separately. The inclusions  $j_i : S \rightarrow S \sqcup S$  send  $s$  to  $s_i$  and identify the sheaf on the  $i$ th copy with  $\mathcal{O}_S$ .

**Proposition 2** (The constructed union is a scheme and a coproduct). *For a scheme  $S$ , the preceding ringed space is a scheme. Together with  $j_0, j_1$ , it is a coproduct in  $\text{Sch}$ : for every scheme  $Y$ , restriction induces a bijection*

$$\begin{aligned} & \text{Hom}(S \sqcup S, Y) \\ & \longrightarrow \text{Hom}(S, Y) \times \text{Hom}(S, Y), \\ & F \longmapsto (Fj_0, Fj_1). \end{aligned}$$

*Proof.* The displayed sheaf is a sheaf because gluing and uniqueness hold on each copy separately. Every point  $s_i$  has an open neighbourhood isomorphic, as a ringed space, to an affine open of  $S$ . Thus the ringed space is locally an affine scheme, proving the first assertion.

Given morphisms  $f_0, f_1 : S \rightarrow Y$ , define  $F$  to equal  $f_i$  on the  $i$ th open copy. There is no overlap between the copies, hence no compatibility condition. The maps on spaces and sheaves assemble uniquely and are locally morphisms of schemes, so they define a unique scheme

morphism  $F$ . Conversely, any  $F$  restricts to these two maps.  $\square$

The symbol  $\sqcup$  is used for the categorical coproduct. The proposition checks that the labelled-copy construction realizes that universal property for schemes; it is not a second, unrelated object. The resulting map  $F$  is often written  $[f_0, f_1]$ , called a *copairing*. Its domain is a union, not a product:

$$\begin{array}{ccccc} S & \xrightarrow{j_0} & S \sqcup S & \xleftarrow{j_1} & S \\ & \searrow f_0 & \downarrow F & \swarrow f_1 & \\ & & Y & & \end{array}$$

For  $Y = S$  and  $f_0 = f_1 = \text{id}_S$ , this constructs the scheme morphism

$$\pi = [\text{id}_S, \text{id}_S] : S \sqcup S \longrightarrow S.$$

Its underlying map is  $s_i \mapsto s$ , and its pull-back of functions on an open  $U \subset S$  is  $a \mapsto (a, a)$  in  $\mathcal{O}_S(U) \times \mathcal{O}_S(U)$ . This specifies both parts of the scheme morphism. Its defining equations are  $\pi j_0 = \pi j_1 = \text{id}_S$ . The notation  $[\text{id}_S, \text{id}_S]$  does not mean a function taking values in ordered pairs; the pair records the two restrictions of a single function. Interchanging labels defines an automorphism  $\sigma(s_i) = s_{1-i}$ , with  $\sigma^2 = \text{id}$  and  $\pi\sigma = \pi$ . The composite  $S \sqcup S \rightarrow S \rightarrow B$  makes the union a  $B$ -scheme, and  $\pi$  is an arrow both in  $\text{Sch}/B$  and in  $\text{Sch}/S$ .

**Example 8** (Two affine lines, with their coordinate rings). Take  $B = \text{Spec } k$ ,  $A =$

$k[z]$ , and  $S = \operatorname{Spec} A = \mathbb{A}_k^1$ . The disjoint union of two affine lines is

$$S \sqcup S \cong \operatorname{Spec}(k[z] \times k[z]).$$

Indeed the idempotents  $(1, 0)$  and  $(0, 1)$  split this spectrum into disjoint open subsets. Localizing at either idempotent gives  $k[z]$ , so these subsets have exactly the structure sheaves of the two affine lines. This is the product-ring calculation in Lemma 10.21.2, Tag 00ED, including the localizations in its proof.

The fold and the two inclusions correspond to these ring maps:

$$\begin{aligned} \pi^* : k[z] &\longrightarrow k[z] \times k[z], \\ p(z) &\longmapsto (p(z), p(z)), \\ j_0^* : k[z] \times k[z] &\longrightarrow k[z], \\ (p(z), q(z)) &\longmapsto p(z), \\ j_1^* : k[z] \times k[z] &\longrightarrow k[z], \\ (p(z), q(z)) &\longmapsto q(z). \end{aligned}$$

For example, the two labelled points above  $z = 2$  have prime ideals  $(z - 2) \times k[z]$  and  $k[z] \times (z - 2)$ . Contracting either ideal along  $\pi^*$  gives  $(z - 2)$  in  $k[z]$ . The involution interchanges the factors of the product ring.

Compare three operations on the same  $S$ :

$$\begin{aligned} S \sqcup S &= \operatorname{Spec}(k[z] \times k[z]), \\ S \times_k S &= \operatorname{Spec} k[z_0, z_1], \\ S \times_S S &\cong \operatorname{Spec} k[z]. \end{aligned}$$

They give two disjoint lines, a plane, and one line, respectively.

The objects in our running example have the form

$$(S, \pi : P \rightarrow S, \sigma).$$

Here  $S$  varies among  $B$ -schemes,  $P$  is a scheme over  $S$ , and  $\sigma$  is the sheet-exchanging involution. We will make “double cover” precise in Section 7: it means a cover that becomes two copies after an allowed base change, not just a map that happens to be two-to-one on some set of points.

The *total* collection lets  $S$  vary; for example it includes the trivial cover of  $\operatorname{Spec} k$  and the trivial cover of  $\mathbb{A}_k^1$  when  $B = \operatorname{Spec} k$ . The collection over one fixed  $S$  only includes covers of that  $S$ . These are the object data of Definition 1 in the running example. With the arrows there specified and the category axioms checked in Section 9, they give  $BG$  and its fiber  $(BG)_S$ , respectively. The fiber keeps only arrows over  $\operatorname{id}_S$ , not every arrow between those objects in the total category.

A *nontrivial* double cover is one not isomorphic over  $S$  to the fold map. One example, justified in Section 6, is  $\operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{R}$  with complex conjugation. This example also warns against counting topological points: each spectrum has one point, but the map is a double cover in the algebraic sense.

### Checkpoint.

- ◇ Defined:  $B$ -schemes,  $\operatorname{Sch}/B$ , categories over a base, and fibers  $\mathcal{F}_S$ .

- ◇ Constructed: labelled disjoint unions and affine fiber products.
- ◇ Proved: two fiber products are uniquely isomorphic preserving their projections.
- ◇ Check: write the fold map on labelled elements and on functions.
- ◇ Check: distinguish a map into a fiber product from an isomorphism between two fiber products.

## 4 Test schemes and the functor of points

### 4.1 A test point is a morphism over the fixed base

**Definition 7** ( $T$ -valued points). Fix  $B$ -schemes  $b_X : X \rightarrow B$  and  $b_T : T \rightarrow B$ . Define the set

$$X(T) := \text{Hom}_B(T, X).$$

An element  $\xi \in X(T)$  is therefore a scheme morphism  $\xi : T \rightarrow X$  satisfying  $b_X \circ \xi = b_T$ . It is called a  $T$ -valued point, or a  $T$ -point, of  $X$ .

The condition “over  $B$ ” is the commutative triangle

$$\begin{array}{ccc} T & \xrightarrow{\xi} & X \\ & \searrow b_T & \swarrow b_X \\ & B & \end{array}$$

We call the source  $T$  a *test scheme* when considering this set of maps into  $X$ . For example, when  $B = \text{Spec } k$ , choosing  $T = \text{Spec } k$  asks for rational points; choosing  $T = \text{Spec } k[\epsilon]/(\epsilon^2)$  allows maps that also detect nilpotent functions, as Example 11 computes.

*Remark 4* (Notation and the suppressed base). The notation  $X(T)$  is standard: it is exactly Poonen’s Definition 2.3.1 in [3, Section 2.3]. The base is suppressed in that notation, but remains part of both objects. Thus  $X(T)$  means maps over our specified  $B$ , not all scheme maps  $T \rightarrow X$ . The longer notation  $h_X(T)$  will denote the same set. The letter  $\xi$  will name a morphism; a letter such as  $z$  will name a polynomial variable. Keeping these roles separate helps in the computations below.

### 4.2 The functor of points: both assignments

**Definition 8** (Functor of points). For a fixed  $B$ -scheme  $X$ , its functor of points is

$$h_X : (\text{Sch}/B)^{\text{op}} \longrightarrow \text{Sets}.$$

On objects, it assigns  $T \mapsto h_X(T) = X(T)$ . On an arrow  $u : T' \rightarrow T$  of  $B$ -schemes, it assigns the function

$$\begin{aligned} h_X(u) : X(T) &\longrightarrow X(T'), \\ \xi &\longmapsto \xi \circ u. \end{aligned}$$

Here Sets is the category of sets and functions, and op records that arrows reverse direction.

This is the functor in Stacks Project Example 4.3.4, Tag 0010, applied to  $\text{Sch}/B$ ; see also Schemes, Section 26.13, Tag 01J5.

Precomposition is pictured by

$$\begin{array}{ccc} T' & \xrightarrow{u} & T \\ & \searrow \xi \circ u & \swarrow \xi \\ & X & \end{array}$$

All arrows are over  $B$ , so the composite is again an allowed test point. The identity law is  $\xi \circ \text{id}_T = \xi$ . If  $v : T'' \rightarrow T'$ , associativity gives

$$(\xi \circ u) \circ v = \xi \circ (u \circ v).$$

Consequently  $h_X(v) \circ h_X(u) = h_X(u \circ v)$ , which is the composition law for this contravariant functor. We have specified the object assignment, the arrow assignment, and checked both functor laws.

### 4.3 Affine test schemes turn the definition into algebra

**Theorem 5** (Affine morphisms and ring maps). *For commutative rings  $A$  and  $R$ , taking the induced map on global sections gives a bijection*

$$\begin{aligned} \text{Hom}_{\text{Sch}}(\text{Spec } R, \text{Spec } A) \\ \xrightarrow{\sim} \text{Hom}_{\text{rings}}(A, R), \\ \xi \longmapsto \xi^\#. \end{aligned}$$

*The inverse sends a ring map  $\theta : A \rightarrow R$  to the scheme morphism  $\text{Spec } \theta : \text{Spec } R \rightarrow \text{Spec } A$ .*

This is the affine case of Lemma 26.6.4, Tag 01I1; Lemma 26.6.5, Tag 01I2, packages it as an equivalence between affine schemes and the opposite category of rings. This is a theorem about schemes, not Yoneda's lemma. For orientation, the underlying map sends a prime ideal  $\mathfrak{q} \subset R$  to  $\theta^{-1}(\mathfrak{q}) \subset A$ ; localization of  $\theta$  gives the required maps on structure sheaves. The theorem asserts that this procedure and passage to global sections undo each other.

Now specialize explicitly to  $B = \text{Spec } k$ , where  $k$  is a field,  $X = \text{Spec } A$ , and  $T = \text{Spec } R$ , with specified  $k$ -algebra structures on  $A$  and  $R$ . Being a map *over*  $\text{Spec } k$  means that the corresponding ring map respects  $k$ . The two triangles are

$$\begin{array}{ccc} \text{Spec } R & \xrightarrow{\xi} & \text{Spec } A \\ & \searrow & \swarrow \\ & \text{Spec } k & \end{array}$$

and, with directions reversed,

$$\begin{array}{ccc} A & \xrightarrow{\xi^\#} & R \\ & \swarrow & \searrow \\ & k & \end{array}$$

Thus the theorem restricts to a bijection

$$X(\text{Spec } R) \simeq \text{Hom}_{k\text{-alg}}(A, R).$$

The abbreviation  $X(R)$  for  $X(\text{Spec } R)$  is also customary when the base ring is understood. It does not mean that a ring has become a scheme without applying  $\text{Spec}$ .

**Example 9** (Recall: rational points). Over  $B = \operatorname{Spec} k$ , a  $k$ -rational point of  $X$  is a  $k$ -morphism  $\xi : \operatorname{Spec} k \rightarrow X$ ; their set is denoted  $X(k)$ . For an affine scheme defined by polynomial equations, this recovers solutions whose coordinates lie in  $k$ . The word “rational” here does not require  $k = \mathbb{Q}$ .

For instance, let  $X = \mathbb{A}_k^1 = \operatorname{Spec} k[z]$ . By Theorem 5, a  $k$ -point corresponds to a  $k$ -algebra map  $k[z] \rightarrow k$ . Such a map is determined by one element  $a \in k$ , the image of  $z$ , and is evaluation:

$$\xi_a^\#(F(z)) = F(a).$$

The single point of  $\operatorname{Spec} k$  maps to the prime ideal  $(z - a)$ . Therefore  $\mathbb{A}_k^1(k) \simeq k$ . For a general affine  $X = \operatorname{Spec} k[z_1, \dots, z_n]/I$ , the same argument gives the tuples  $(a_1, \dots, a_n) \in k^n$  annihilating every polynomial in  $I$ . This is the coordinate interpretation recalled in [3, Section 2.3.1].

**Example 10** (An affine line has points valued in any algebra). Keep  $B = \operatorname{Spec} k$  and  $X = \operatorname{Spec} k[z]$ , but now take  $T = \operatorname{Spec} R$  for any  $k$ -algebra  $R$ . Then

$$X(T) \simeq \operatorname{Hom}_{k\text{-alg}}(k[z], R) \simeq R.$$

The second bijection sends  $\xi^\#$  to  $\xi^\#(z)$ . Conversely,  $r \in R$  gives evaluation  $F(z) \mapsto F(r)$ , so there is no condition on  $r$ . For example, with  $R = k[t]$ , the rule  $z \mapsto t^2$  determines a  $T$ -point  $\xi : T \rightarrow X$ . This is a morphism from an entire affine line, not a choice of a single topological point of  $X$ .

**Example 11** (Dual numbers detect what field-valued points miss). Let  $B = \operatorname{Spec} k$  and  $X = \operatorname{Spec} k[z]/(z^2)$ . A test point with source  $T = \operatorname{Spec} R$  corresponds to a  $k$ -algebra map

$$\xi^\# : k[z]/(z^2) \longrightarrow R.$$

Its value  $r = \xi^\#(\bar{z})$  must satisfy  $r^2 = 0$ ; conversely, every such  $r$  gives a map. Thus

$$X(\operatorname{Spec} R) \simeq \{r \in R : r^2 = 0\}.$$

If  $R = L$  is a field extension of  $k$ , the only possibility is  $r = 0$ . Field-valued points cannot distinguish  $X$  from  $\operatorname{Spec} k$  in this example: each has exactly one  $L$ -point for every such  $L$ .

Instead use the *dual numbers*  $R = k[\epsilon]/(\epsilon^2)$ , where  $\epsilon$  denotes the residue class and is nonzero although  $\epsilon^2 = 0$ . An element is uniquely  $a + b\epsilon$  with  $a, b \in k$ . Its square vanishes precisely when  $a = 0$ . Therefore

$$X(\operatorname{Spec} R) \simeq \{b\epsilon : b \in k\}.$$

In particular, the assignments  $\bar{z} \mapsto 0$  and  $\bar{z} \mapsto \epsilon$  give two different scheme morphisms  $\xi_0, \xi_1 : T \rightarrow X$ . Their underlying maps of topological spaces are the same: both spaces have just one point. They differ on functions, since  $\xi_0^\#(\bar{z}) = 0$  and  $\xi_1^\#(\bar{z}) = \epsilon$ . Allowing nonreduced test schemes detects this difference. It does not manufacture extra topological points.

#### 4.4 Changing the target: naturality as a square

Keep  $X, Y$  over  $B$ , and let  $f : X \rightarrow Y$  be a  $B$ -morphism. For every test scheme  $T$ , postcomposition gives a function

$$\begin{aligned} h_f(T) : X(T) &\longrightarrow Y(T), \\ \xi &\longmapsto f \circ \xi. \end{aligned}$$

This time the arrow  $f : X \rightarrow Y$  and the induced function point in the same direction; the reversal in  $h_X$  concerned its source  $T$ . For each  $u : T' \rightarrow T$ , there is a commutative square of *sets*

$$\begin{array}{ccc} X(T) & \xrightarrow{h_f(T)} & Y(T) \\ h_X(u) \downarrow & & \downarrow h_Y(u) \\ X(T') & \xrightarrow{h_f(T')} & Y(T') \end{array}$$

Starting with  $\xi \in X(T)$ , the two routes give respectively  $(f \circ \xi) \circ u$  and  $f \circ (\xi \circ u)$ , which agree by associativity. Equivalently,

$$h_f(T')(\xi \circ u) = h_f(T)(\xi) \circ u.$$

A collection of functions  $X(T) \rightarrow Y(T)$  is called a *natural transformation* when it commutes with precomposition in this way for every  $u$ . Thus “natural in  $T$ ” means this explicit compatibility, not an unspecified property of the formulas.

#### 4.5 The Yoneda argument we will actually use

**Theorem 6** (Yoneda for the functors of points). *Suppose functions  $\eta_T : X(T) \rightarrow$*

*$Y(T)$  are given for every  $B$ -scheme  $T$ , and for every  $u : T' \rightarrow T$  satisfy*

$$\eta_{T'}(\xi \circ u) = \eta_T(\xi) \circ u.$$

*There is a unique  $B$ -morphism  $f : X \rightarrow Y$  such that  $\eta_T(\xi) = f \circ \xi$  for every  $T$  and every  $\xi \in X(T)$ .*

This is Lemma 4.3.5, Tag 001P, applied to the category  $\text{Sch}/B$ . Here the short proof is useful enough to give in full.

*Proof.* Use  $X$  itself as the test scheme. Its identity morphism is an element of  $X(X)$ . Define

$$f := \eta_X(\text{id}_X) \in Y(X) = \text{Hom}_B(X, Y).$$

This is already a genuine  $B$ -morphism by the definition of  $Y(X)$ . We must show it gives all the prescribed functions, not just the one at  $X$ . For any  $\xi : T \rightarrow X$ , naturality gives the square

$$\begin{array}{ccc} X(X) & \xrightarrow{\eta_X} & Y(X) \\ h_X(\xi) \downarrow & & \downarrow h_Y(\xi) \\ X(T) & \xrightarrow{\eta_T} & Y(T). \end{array}$$

Apply it to  $\text{id}_X$  in the upper left. The left-then-bottom route gives  $\eta_T(\text{id}_X \circ \xi) = \eta_T(\xi)$ ; the top-then-right route gives  $f \circ \xi$ . This proves existence. If  $f'$  induced the same functions, evaluation at  $\text{id}_X$  would give  $f' = f' \circ \text{id}_X = \eta_X(\text{id}_X) = f$ . This proves uniqueness.  $\square$

*Remark 7* (How pointwise notation will be used). In these notes the formula  $(g, p) \mapsto g \cdot p$  means a rule for maps from every  $T$ , compatible with precomposition. It does not mean a rule only on the underlying topological points. Yoneda then turns such a natural rule into a unique scheme morphism. To check equality of two already defined scheme maps, it likewise suffices to check every test scheme: using their source and its identity point detects equality.

For fiber products the universal property gives

$$(T \times_S X)(W) \simeq T(W) \times_{S(W)} X(W).$$

On the right is a fiber product of *sets of morphisms*: its elements are compatible pairs of maps from  $W$ . It is not the product of the underlying topological spaces of the schemes.

### Checkpoint.

- ◇ Defined  
 $X(T) = h_X(T) = \text{Hom}_B(T, X)$ ,  
 with its base triangle.
- ◇ Defined  $h_X$  on objects and arrows;  
 verified its functor laws.
- ◇ Computed rational, affine-line, and  
 dual-number test points.
- ◇ Expressed naturality by a  
 commutative square and proved  
 the Yoneda recovery formula  
 $f = \eta_X(\text{id}_X)$ .

## 5 Group schemes, built from examples

### 5.1 An ordinary group is not automatically a scheme

Let  $H = \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$  be the abstract two-element group. This is a set with an addition law. We do *not* take a map from this bare two-element set into  $B$  and call it a group scheme. Instead we build an actual scheme

$$H_B = B_0 \amalg B_1 \longrightarrow B,$$

where each  $B_a$  is a copy of  $B$  and the map is the fold map of Example 8. Over every point of  $B$  there is a copy with label 0 and a copy with label 1. If  $B$  has infinitely many points,  $H_B$  does too.

The labels tell us how to define a multiplication morphism. There are four components in  $H_B \times_B H_B$ , indexed by  $(a, b) \in H^2$ . Each is a copy of  $B$ . Map it by the identity on  $B$  to the component of  $H_B$  labeled  $a + b$  modulo 2:

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| $(a, b)$ | $(0, 0)$ | $(0, 1)$ | $(1, 0)$ | $(1, 1)$ |
| $a + b$  | 0        | 1        | 1        | 0.       |

The unit map  $e : B \rightarrow H_B$  is inclusion of  $B_0$ . Inversion is the identity on both components because  $-a = a$  in  $H$ . These are genuine morphisms of schemes, defined component by component. This is our first group scheme.

## 5.2 What “group object” packages

An ordinary group can be described by a multiplication  $H \times H \rightarrow H$ , a unit map  $\{*\} \rightarrow H$  selecting the identity, and an inverse map  $H \rightarrow H$ . The group axioms are equations between composites of these maps. To make a group object in  $\text{Sch}/B$ , replace sets by  $B$ -schemes, products by fiber products over  $B$ , and the one-point terminal set by the terminal  $B$ -scheme  $(B, \text{id}_B)$ .

**Definition 9** (Group scheme). A group scheme over  $B$  is a  $B$ -scheme  $\pi_G : G \rightarrow B$  equipped with  $B$ -morphisms

$$\begin{aligned} m &: G \times_B G \rightarrow G, \\ e &: B \rightarrow G, \\ \iota &: G \rightarrow G \end{aligned}$$

for multiplication, unit, and inverse. They satisfy associativity, the two unit laws, and the two inverse laws.

For precision, write  $G^n$  for the  $n$ -fold product over  $B$ . Associativity is the commutative square

$$\begin{array}{ccc} G^3 & \xrightarrow{m \times \text{id}_G} & G^2 \\ \text{id}_G \times m \downarrow & & \downarrow m \\ G^2 & \xrightarrow{m} & G \end{array}$$

The unit and inverse laws, with each pairing a map  $G \rightarrow G^2$ , are

$$\begin{aligned} m \langle e\pi_G, \text{id}_G \rangle &= \text{id}_G, \\ m \langle \text{id}_G, e\pi_G \rangle &= \text{id}_G, \\ m \langle \iota, \text{id}_G \rangle &= e\pi_G, \\ m \langle \text{id}_G, \iota \rangle &= e\pi_G. \end{aligned}$$

The phrase “over  $B$ ” includes  $\pi_G e = \text{id}_B$  and, for example,  $\pi_G m = \pi_G \text{pr}_1 = \pi_G \text{pr}_2$ . Multiplication combines two elements over the *same* base; it does not combine unrelated fibers. There need not be a group structure on the underlying topological set of  $G$ .

For every  $T \rightarrow B$ , the set  $G(T) = \text{Hom}_B(T, G)$  is an ordinary group: multiply  $g, h : T \rightarrow G$  by  $m \circ (g, h)$ ; its identity is  $T \rightarrow B \xrightarrow{e} G$ ; its inverse sends  $g$  to  $\iota g$ . The diagram equations give the usual group axioms on this set. This is equivalent to Definition 39.4.1, Tag 022S; the equivalence uses precisely the Yoneda argument of Section 4.

For a finite abstract group  $H$ , the same component construction gives  $H_B = \coprod_{h \in H} B$  with products of labels given by the group law. Its  $T$ -points are locally constant maps  $T \rightarrow H$ : the inverse images of the labeled components are an open-and-closed partition of  $T$ . If  $T$  is nonempty and connected, this is just a choice of one element of  $H$ . For disconnected  $T$  there can be more choices. This construction is called the *constant group scheme* (Example 39.5.6 in Tag 047F).

## 5.3 An infinite example: addition on the affine line

Let  $B = \text{Spec } A$ . Take  $G = \text{Spec } A[x]$ , and define multiplication by the ring map

$$A[x] \longrightarrow A[x_1, x_2], \quad x \longmapsto x_1 + x_2.$$

The unit is given by  $x \mapsto 0$  in  $A$ ; inversion by  $x \mapsto -x$ . Their equations are the ordinary addition laws. This group scheme is denoted  $\mathbb{G}_{a,B}$ . On  $T = \operatorname{Spec} R$  it gives the additive group  $(R, +)$ ; in particular  $\mathbb{G}_{a,k}(k) = (k, +)$ , an infinite group when  $k$  is infinite. We have put a group law on a familiar scheme, not replaced a group by an unrelated topological space.

Likewise  $\mathbb{G}_{m,B} = \operatorname{Spec} A[x, x^{-1}]$  uses  $x \mapsto x_1 x_2$  for multiplication,  $x \mapsto 1$  for its unit, and  $x \mapsto x^{-1}$  for inversion. Its  $R$ -points form the group  $R^\times$ . These constructions glue over arbitrary  $B$ ; see Examples 39.5.1 and 39.5.3 in Tag 047F.

#### 5.4 Changing the base of a group scheme

For a specified  $s : S \rightarrow B$ , form

$$\begin{array}{ccc} G_S = G \times_B S & \longrightarrow & G \\ \pi_{G_S} \downarrow & & \downarrow \pi_G \\ S & \xrightarrow{s} & B \end{array}$$

The result is an  $S$ -scheme. It becomes a group scheme over  $S$  by pulling back the three structure maps. For example,

$$G_S \times_S G_S \simeq (G \times_B G) \times_B S,$$

so  $m_S$  is  $m$  pulled back to  $S$ . On test points the formulas are

$$\begin{aligned} m_S((g_1, t), (g_2, t)) &= (m(g_1, g_2), t), \\ e_S(t) &= (e(s(t)), t), \\ \iota_S(g, t) &= (\iota(g), t). \end{aligned}$$

The shared  $t$  is exactly the condition in the product over  $S$ . The original equations pull back to the new equations, so these maps give a group scheme (Lemma 39.4.2, Tag 022T). Thus  $G_S$  is not a single ordinary group: it is a scheme over  $S$  with a group law, and  $G_S(T)$  is an ordinary group for every  $T \rightarrow S$ . For  $H_B = B \amalg B$  this construction gives  $H_S = S \amalg S$  with the same label addition. For  $\mathbb{G}_{a,B}$  it gives  $\mathbb{A}_S^1$  with addition.

#### 5.5 The geometric hypotheses

These adjectives apply to the *scheme morphism*  $\pi_G : G \rightarrow B$ , not to an abstract set of group elements. Recall the following definitions for any scheme morphism  $f : X \rightarrow Y$ .

*Affine.* The inverse image of each affine open  $\operatorname{Spec} A \subset Y$  is an affine scheme  $\operatorname{Spec} C$ . This is a relative property;  $Y$  need not itself be affine (Section 29.11, Tag 01S5).

*Locally of finite presentation.* Around every point of  $X$  we can choose affine opens  $\operatorname{Spec} C \subset X$  and  $\operatorname{Spec} A \subset Y$  with the first mapping into the second, such that

$$C \simeq A[x_1, \dots, x_n]/(f_1, \dots, f_r)$$

as an  $A$ -algebra, for finite  $n, r$ . The coefficients belong to the base ring  $A$ , not necessarily  $\mathbb{C}$ . Thus the recollection “finitely many variables and a finitely generated ideal of relations” is exactly right. It is not a claim that the scheme has finitely many points or that its group of sections

is a finitely presented abstract group. Finite presentation of a morphism without “locally” additionally requires quasi-compactness and quasi-separatedness (Definition 29.22.1, Tag 01TP). These additional global conditions are not prerequisites for the torsor proof; the local condition is the one imposed on  $G \rightarrow B$ .

*Flat.* A ring map  $A \rightarrow C$  is flat if tensoring an exact sequence of  $A$ -modules with  $C$  preserves exactness. Since tensor product is already right exact, this means it also preserves injections. A scheme morphism is flat when its induced local-ring maps are flat; one can equivalently check the corresponding ring maps on affine charts (Section 29.26, Tag 01U2). This controls how algebraic relations behave in a family; it is not a synonym for smoothness. Free modules are flat, which verifies all the examples needed immediately.

Check these conditions for  $H_B$  with  $H = \mathbb{Z}/2\mathbb{Z}$ . Above  $\text{Spec } A$  its algebra is

$$A \times A \simeq A[z]/(z^2 - z).$$

Evaluation at  $z = 0, 1$  gives the isomorphism. The two factors are separated by the comaximal ideals  $(z)$  and  $(z - 1)$ , even in characteristic 2. It is affine, has one generator and one relation, and is free of rank two as an  $A$ -module. Hence its map to  $B$  is affine, flat, and locally of finite presentation. For  $\mathbb{G}_{a,B}$ , the algebra  $A[x]$  has one generator and no relations, and is free as an  $A$ -module with basis  $1, x, x^2, \dots$ . This example shows that finite presentation as

an *algebra* does not mean finite generation as a *module*.

**Definition 10** (Standing geometric assumptions). From now on  $G \rightarrow B$  is a group scheme whose structure morphism is affine, flat, and locally of finite presentation.

Its unit section makes it surjective:  $\pi_G e = \text{id}_B$ , so every base point has the point supplied by  $e$  above it. Affineness will let us construct glued torsors using algebras. Flatness and local finite presentation will make torsor projections allowable covering maps. These are sufficient working hypotheses, not a claim of minimality.

### Checkpoint.

- ◇ Constructed:  $(\mathbb{Z}/2\mathbb{Z})_B$ ,  $\mathbb{G}_a$ , and  $\mathbb{G}_m$ .
- ◇ Defined: group-scheme structure maps and the three geometric hypotheses.
- ◇ Verified: the constant group satisfies these hypotheses.
- ◇ Check: distinguish  $H$ ,  $H_B$ , and  $H_B(T)$ .

## 6 Coverings, sites, and sheaves

### 6.1 Start with ordinary open covers

On a topological space, an open cover lets us work on smaller pieces and compare

answers on intersections. For example, continuous functions on  $U_i$  agreeing on  $U_i \cap U_j$  give one continuous function on  $\bigcup_i U_i$ . A site keeps this idea but allows the maps used as local pieces to be more general than open inclusions. It is not another scheme, variety, or topological space.

In these notes a *site* means a category together with a rule specifying which families of arrows  $\{U_i \rightarrow U\}$  count as covers. The covering-family formulation requires three rules:

- (1) An isomorphism, by itself, is a cover.
- (2) Covers of covering pieces combine to give a cover of the original object.
- (3) The fiber products needed to pull a cover back along any arrow exist, and the resulting family covers the new base.

This is Definition 7.6.2, Tag 00VH. The covering rule is often called a Grothendieck topology; formally these families specify a basis for that topology. We need only the families and these rules.

Thus  $(\text{Sch}/B)_{\text{fppf}}$  means precisely “the category of  $B$ -schemes, equipped with the fppf covering rule.” Its objects and arrows are still the objects and arrows of  $\text{Sch}/B$ . In particular, *not every arrow must be fppf*; the extra rule only singles out covering families. Nor have we changed the Zariski topology inside each individual scheme. When this is called a *big site*, the

word indicates that arbitrary  $B$ -schemes are admitted as objects, not merely open subsets of  $B$ . No set-theoretic construction of a “small model” is needed to follow this proof.

## 6.2 The fppf covering rule

**Definition 11.** An fppf cover of  $S$  is a family of  $B$ -morphisms  $\{f_i : S_i \rightarrow S\}$  such that each  $f_i$  is flat and locally of finite presentation, and their images together cover  $S$ .

This is Definition 34.7.1, Tag 021M. The letters abbreviate the French words for faithfully flat and of finite presentation. For a family the surjectivity requirement is *joint*: an individual  $S_i$  need not cover all of  $S$ . A single surjective flat morphism locally of finite presentation is a cover by itself.

An ordinary Zariski open covering is an fppf covering, but the allowed maps need not be open immersions. For example,

$$U = \text{Spec } \mathbb{C} \longrightarrow S = \text{Spec } \mathbb{R}$$

is an fppf cover: the algebra  $\mathbb{C} = \mathbb{R}[z]/(z^2 + 1)$  is free of rank two over  $\mathbb{R}$  and finitely presented, and the map is surjective. It is not an isomorphism of schemes. Over the new base  $U$ , the cover itself becomes

$$U \times_S U \simeq U \amalg U.$$

Indeed, as a  $\mathbb{C}$ -algebra using the first tensor factor,

$$\begin{aligned}\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} &\simeq \mathbb{C}[z]/(z^2 + 1) \\ &\simeq \mathbb{C} \times \mathbb{C},\end{aligned}$$

since  $z^2 + 1 = (z - i)(z + i)$  has distinct roots. Conjugating the second factor exchanges the two components. This is a double cover that splits after an allowed base change. It is not split over  $\mathbb{R}$ : a section would give an  $\mathbb{R}$ -algebra map  $\mathbb{C} \rightarrow \mathbb{R}$ , impossible because  $-1$  has no square root in  $\mathbb{R}$ . Also,  $\text{Spec } \mathbb{R}$  has no smaller nonempty open on which it could split. This is the practical reason to allow more than Zariski covers.

### 6.3 What stability and composition say

*Stable under base change* means the following concrete operation preserves the covering property. If  $\{S_i \rightarrow S\}$  is a cover and  $f : T \rightarrow S$  is *any*  $B$ -morphism, form

$$\begin{array}{ccc} T \times_S S_i & \longrightarrow & S_i \\ \downarrow & & \downarrow \\ T & \xrightarrow{f} & S. \end{array}$$

The conclusion is that  $\{T \times_S S_i \rightarrow T\}_i$  is again an fppf cover. Flatness and local finite presentation survive this pullback, and the pulled-back family is jointly surjective. Geometrically, a point of  $T$  has image in some  $S_i \rightarrow S$  and has a point above it in the corresponding fiber product, allowing a residue-field extension if

necessary. Thus the local pieces still cover after changing base.

*Composition of covers* means exactly covering the covering pieces. If  $\{S_i \rightarrow S\}_i$  is a cover and for each  $i$  we have a cover  $\{U_{ij} \rightarrow S_i\}_{j \in J_i}$ , then the composite family

$$\{U_{ij} \rightarrow S_i \rightarrow S\}_{i,j}$$

covers  $S$ . Composites remain flat and locally of finite presentation; joint surjectivity follows in two steps. The index set  $J_i$  is allowed to depend on  $i$ .

The exact reference is Lemma 34.7.3, Tag 0210: part (1) concerns isomorphisms, part (2) composition, and part (3) base change. In contrast, Lemma 34.7.2, Tag 021N, compares types of covers: it says that syntomic covers, and hence smooth, *étale*, and Zariski covers, are fppf. It does *not* establish the preceding two stability statements. In particular the inclusion is not a statement that every fppf cover is smooth or *étale*.

These verifications are needed even though our initial covers are fppf: pulling back a torsor will produce a *new* covering family, and gluing locally trivial objects will produce a cover of a cover. The lemma ensures those new families are still allowed.

### 6.4 Locality and the use of fpqc theorems

“After this cover” means after pullback along every  $S_i \rightarrow S$ . For a scheme  $P \rightarrow S$ , write  $P_{S_i} = P \times_S S_i$ . Thus “ $P$  is locally isomorphic to  $Q$ ” means  $P_{S_i} \simeq Q_{S_i}$  for some

cover. It need not mean an isomorphism on Zariski open subsets of  $S$ .

We shall occasionally quote theorems stated for *fpqc* covers, a broader class of flat covers that permits maps not locally of finite presentation. In the covering-family definition, every affine open in the target must equal the union of the images of finitely many affine opens in members of the family. Affine opens are quasi-compact: a quasi-compact space is one in which every open cover has a finite subcover. This condition provides the finiteness needed in descent without imposing finitely many algebra generators. This is Definition 34.9.1, Tag 022B.

The fact we use is Lemma 34.9.7, Tag 022C: every fppf cover is an fpqc cover. Consequently an fpqc descent theorem applies to each of *our fppf covers*. We are not changing topology in the proof. Just as a theorem for all groups can be applied to finite groups, a theorem for the larger class can be used on this subclass.

Two further phrases must be distinguished. A property is *preserved by base change* if a map having it still has it after any change of base. It is *detected by a cover* if knowing it for all pulled-back maps implies it for the given original map. For example, if a given map  $P \rightarrow Q$  becomes an isomorphism after an fppf cover of  $S$ , a locality theorem says it was already an isomorphism. The original  $P, Q$  must already exist for this assertion. “Fpqc-local on the target” packages preservation along such covers and detection on

them; it does not by itself assert preservation under *arbitrary*, possibly nonflat, base change.

## 6.5 Presheaves and sheaves: what exactly gets glued?

A *presheaf of sets* assigns a set  $F(T)$  to each object  $T$  and a restriction function  $F(T) \rightarrow F(T')$  to each arrow  $u : T' \rightarrow T$ . Write  $u^*s$  for the restriction of  $s$ . The laws are

$$\text{id}^*s = s, \quad v^*(u^*s) = (uv)^*s.$$

This is the same reversal of arrows as for the functor of points. It does not yet assert that locally defined elements glue.

A presheaf is a *sheaf* for our covering rule if, whenever  $\{T_i \rightarrow T\}$  is a cover and  $s_i \in F(T_i)$  have equal restrictions to every  $T_i \times_T T_j$ , there exists exactly one  $s \in F(T)$  restricting to all  $s_i$ . For open covers this is the familiar agreement-on-intersections rule. Fiber products replace intersections because the new covering pieces need not be subsets.

For example,  $F(T)$  might be functions on  $T$ , or maps from  $T$  to a fixed scheme, or isomorphisms between two objects pulled back to  $T$ . In the last case the *elements being glued are maps*. Later the second stack condition will glue the *objects themselves*; that is a separate problem.

A map of presheaves is a family of functions on  $F(T)$  commuting with restrictions, just as in Section 4. A presheaf  $F$  is *represented by a scheme*  $X$  if there are

compatible bijections  $F(T) \simeq \text{Hom}(T, X)$  for every  $T$  in the base category. This says that the rule  $F$  comes from an actual geometric object  $X$ . Not every rule comes with such an object.

### Checkpoint.

- ◊ Defined: site, fppf cover, presheaf, sheaf, and representability.
- ◊ Verified: fppf covers survive pullback and composition.
- ◊ Check: distinguish an fppf cover from an arbitrary arrow of the site.
- ◊ Check: distinguish gluing maps from gluing objects.

## 7 Torsors: groups without a chosen origin

### 7.1 Actions, sections, and triviality

For an ordinary group  $H$ , a nonempty set on which  $H$  acts freely and transitively looks like  $H$  after one chooses an element to be its origin. Without that choice it is a torsor, not a group with a specified identity. Our scheme definition makes this description work in families.

Given a scheme  $P \rightarrow S$ , an action of  $G_S$  is an  $S$ -morphism  $a_P : G_S \times_S P \rightarrow P$  satisfying  $a_P(e, p) = p$  and  $a_P(g, a_P(h, p)) = a_P(gh, p)$  on all test points. Equivalently these are equations of scheme morphisms. An equivariant map respects this action: applying  $g$  and

then the map is the same as applying the map and then  $g$ . See Definition 39.10.1, Tag 022Z.

A section of  $\pi_P : P \rightarrow S$  is a morphism  $s : S \rightarrow P$  with  $\pi_P s = \text{id}_S$ . It chooses an element over every base point in a way that is a morphism of schemes, not an arbitrary set-theoretic choice. For a double cover with sheet swap, such a section labels its sheet as 0 and the other sheet as 1. A torsor need not have a global section, but it must have such descriptions after allowed base changes.

### 7.2 The definition with its local isomorphisms

**Definition 12.** A left  $G_S$ -torsor is a scheme  $\pi_P : P \rightarrow S$  with an action

$$a_P : G_S \times_S P \longrightarrow P$$

and an fppf cover  $\{U_a \rightarrow S\}$  on which there are equivariant isomorphisms  $P_{U_a} \simeq G_{U_a}$ . On  $G_{U_a}$  the action is left multiplication. The action must satisfy the unit and associativity laws.

Our convention agrees with Definition 39.11.3, Tag 049A. Trivializations are required to exist; they are not extra chosen data in the definition of the object.

The associated *torsor identity* is the map

$$\begin{aligned} \Theta_P : G_S \times_S P &\longrightarrow P \times_S P, \\ (g, p) &\longmapsto (g \cdot p, p). \end{aligned} \quad (7.1)$$

It is an isomorphism. On a trivialization its inverse is  $(p_1, p_2) \mapsto (p_1 p_2^{-1}, p_2)$ , and

being an isomorphism is fpqc-local on the base (Lemma 35.23.19, Tag 02L4).

In particular, this asserts simple transitivity on sections after any test-scheme base change whenever a section is available. It is stronger than a statement only about geometric points.

**Example 12** (The torsor identity on four copies). Take  $G = (\mathbb{Z}/2\mathbb{Z})_B$  and the trivial torsor  $P = S_0 \sqcup S_1$ , where the subscripts label the two copies of  $S$ . Both  $G_S \times_S P$  and  $P \times_S P$  consist of four copies of  $S$ . The map  $\Theta_P$  acts on their labels as follows:

| source $(g, p)$ | target $(g + p, p)$ |
|-----------------|---------------------|
| $(0, 0)$        | $(0, 0)$            |
| $(1, 0)$        | $(1, 0)$            |
| $(0, 1)$        | $(1, 1)$            |
| $(1, 1)$        | $(0, 1)$            |

On each labeled copy it is the identity map of  $S$ . Thus this table describes an isomorphism of schemes, not merely a bijection between four abstract points. Given target labels  $(a, b)$ , the unique group label carrying  $b$  to  $a$  is  $g = a - b$  in  $\mathbb{Z}/2\mathbb{Z}$ . The inverse sends  $(a, b)$  to  $(a - b, b)$  and again preserves the base. This is what “there is a unique group element taking the second point to the first” looks like as a scheme morphism.

**Proposition 3.** *Under the standing assumptions on  $G$ , Definition 12 is equivalent to requiring that  $P \rightarrow S$  be surjective, flat, and locally of finite presentation and that  $\Theta_P$  be an isomorphism.*

*Proof.* Local triviality proves the identity as above and transfers flatness and local finite presentation from  $G_S \rightarrow S$  to  $P \rightarrow S$ . These properties are fpqc-local on the base (Lemmas 35.23.17 and 35.23.13, Tag 02L2 and Tag 02KY). Surjectivity follows on the same trivializing cover, since  $G_S$  has a unit section.

Conversely, use the singleton fppf cover  $P \rightarrow S$ . The pulled-back object  $P \times_S P \rightarrow P$ , with projection to its second factor, is equivariantly isomorphic via  $\Theta_P$  to  $G_S \times_S P \rightarrow P$ . It is therefore trivial on this cover.  $\square$

More generally, a section  $s : S \rightarrow P$  trivializes a torsor by  $g \mapsto g \cdot s$ . Its inverse sends  $p$  to the unique  $g$  with  $p = g \cdot s(\pi_P(p))$ , furnished by  $\Theta_P^{-1}$ . This is the section criterion of Lemma 39.11.2, Tag 0499.

### 7.3 Sheaf torsors and representability

The proof below can be followed using schemes as torsors throughout. This subsection explains a different definition often encountered in references. It separates a rule for local sections from the question whether a scheme realizes that rule. The representability paragraph previews the gluing theorem of Section 16; its “cocycle” means the consistency equation for three local identifications, defined fully in Section 14.

For a scheme  $X \rightarrow S$ , write  $h_X(T) = \text{Hom}_S(T, X)$ . A topology is *subcanonical* if each such representable functor is a

sheaf. The fppf topology is subcanonical: schemes satisfy the stronger fpqc sheaf condition (Lemma 35.13.7, Tag 023Q).

A torsor under the sheaf of groups  $h_{G_S}$  is a sheaf of sets  $\mathcal{P}$  on  $(\text{Sch}/S)_{\text{fppf}}$ , with an action, locally isomorphic to  $h_{G_S}$  with its regular action. No scheme is part of this definition. For an already represented sheaf  $h_P$ , the two torsor definitions agree (Lemma 39.11.4, Tag 049B).

For affine  $G$ , every such sheaf torsor is represented by an affine  $S$ -scheme. Here is the representability argument and its precise input. Choose local trivializations by affine  $U_a$ -schemes  $G_{U_a}$ . Yoneda turns their sheaf transition maps into scheme isomorphisms satisfying a cocycle. Effective descent for affine morphisms, stated in Theorem 10, produces an affine  $S$ -scheme  $P$ . The local identifications  $h_P|_{U_a} \simeq \mathcal{P}|_{U_a}$  glue to  $h_P \simeq \mathcal{P}$  by the sheaf axiom. Thus representability uses affine descent, not the claim that  $BG$  is a stack; compare Remark 39.11.5, Tag 049C. One can also allow geometric objects called algebraic spaces, which are needed for the comparison with curves later. No knowledge of them is needed in the torsor proof: affineness lets us stay inside schemes.

**Example 13** (The first double cover). Let  $G = (\mathbb{Z}/2\mathbb{Z})_B = B \amalg B$ . Then

$$G_S \times_S S \simeq G_S = S \amalg S \rightarrow S$$

is the trivial torsor. The nonidentity element exchanges the two sheets. This

group scheme is finite *étale* over every base. Its sections over  $T$  are locally constant maps from  $T$  to  $\mathbb{Z}/2\mathbb{Z}$ .

On an affine base  $\text{Spec } A$ ,

$$\mu_2 = \text{Spec } A[z]/(z^2 - 1).$$

If 2 is invertible, the Chinese remainder theorem identifies it with  $\text{Spec}(A \times A)$ . Over a nonempty  $\mathbb{F}_2$ -scheme it has a nonzero square-zero coordinate  $z - 1$ , whereas the constant group has two disjoint copies of the base and is *étale*. The two groups are therefore different in characteristic 2. This distinction is relative to the base, which itself might be nonreduced. Over a characteristic-two field-valued base point, their fibers are respectively  $\text{Spec } k[\epsilon]/(\epsilon^2)$  and  $\text{Spec } k \amalg \text{Spec } k$ , so they cannot be isomorphic over that base.

**Exercise 2** (The torsor identity alone). Let  $S \neq \emptyset$ . Show that  $P = \emptyset$  satisfies  $G_S \times_S P \simeq P \times_S P$  but is not a torsor. Identify the missing condition.

**Exercise 3** (Sections and triviality). Prove that a torsor  $P \rightarrow S$  is trivial if and only if it has a section. Describe how changing a section changes the trivialization.

**Exercise 4** (Two groups of order two). Compare the coordinate algebras of  $(\mathbb{Z}/2\mathbb{Z})_{\text{Spec } k}$  and  $\mu_{2,k}$  when  $\text{char } k = 2$ . What are their  $k$ -points? Explain why a comparison only of orders or ranks cannot identify the group schemes.

### 7.4 The topology choice revisited

The single fppf cover  $\mathrm{Spec} \mathbb{C} \rightarrow \mathrm{Spec} \mathbb{R}$  already showed why Zariski local triviality is too restrictive. For the constant group of order two, the torsors are finite *étale* double covers; this means in particular that they split on *étale* covers, a class of algebraic local coordinate changes broader than open immersions. The precise splitting theorem is used in Section 18. All these covers are fppf, so we can keep one covering rule throughout.

For reference, if  $G \rightarrow B$  is smooth, an fppf torsor is smooth and surjective over its base: smoothness is detected on the trivializing cover (Lemma 35.23.29, Tag 02VL). Smooth surjective morphisms have *étale* local sections (Lemma 37.38.6, Tag 055U), so such torsors are already *étale*-locally trivial. The fppf rule also accommodates nonsmooth groups such as  $\mu_2$  in characteristic 2. Under our flatness and presentation hypotheses, fpqc-local triviality gives the same torsors (Lemma 95.15.4, Tag 0CQJ). These comparisons are background; the proof uses fppf covers and the explicitly stated descent theorems only.

#### Checkpoint.

- ◇ Defined: action, equivariance, section, torsor, and local triviality.
- ◇ Verified: a section trivializes a torsor; the torsor identity expresses simple transitivity.

- ◇ Check: explain why a double cover can be locally trivial without a global choice of sheet.

**Theorem 8** (Target theorem). *Let  $B$  be a scheme and let  $G \rightarrow B$  satisfy Definition 10. The category of scheme-represented fppf  $G$ -torsors, with equivariant arrows over varying  $B$ -schemes, is a stack in groupoids on  $(\mathrm{Sch}/B)_{\mathrm{fppf}}$ . It realizes*

$$BG = [B/G] = [\mathrm{pt}/G],$$

where  $\mathrm{pt}$  is the relative point  $B$ .

This is the assertion to be proved, not an assumption. The next sections first construct the category and then verify each of the properties in the roadmap.

## 8 The meaning of the quotient notation

Before constructing the total category, distinguish four meanings of a quotient by a group.

First consider an ordinary group  $H$ . Make a category with one object, called  $*$ , and one arrow  $* \rightarrow *$  for each element of  $H$ . Compose arrows using multiplication in  $H$ ; the identity element gives the identity arrow, and  $h^{-1}$  gives the inverse of the arrow  $h$ . A category whose arrows are all invertible is called a *groupoid*. An invertible arrow from an object to itself is an *automorphism*. Thus this one-object groupoid remembers the whole group of symmetries, although its set of objects has only one member.

For a  $B$ -scheme  $S$ , we can do this with the ordinary group  $G(S) = \mathrm{Hom}_B(S, G)$ . The important point is that changing  $S$  changes the group of sections. We must retain this dependence on  $S$ , not just choose one group once and for all.

*The action groupoid.* The trivial action on the relative point is encoded by the diagram  $G \rightrightarrows B$ : both arrows are the structure map, while multiplication in  $G$  specifies composition. This is called an *internal groupoid* because its objects and arrows are themselves specified by schemes. For each  $B$ -scheme  $S$ , it gives the one-object groupoid with automorphism group  $G(S)$  just described. It retains the stabilizer, meaning the group of symmetries of the object; the set of orbits would be a singleton.

*The naive quotient prestack.* Varying  $S$  gives a groupoid-valued presheaf  $\mathcal{B}_0$  with one object over  $S$  and automorphisms  $G(S)$ . Pullback of arrows is restriction of sections. Because  $G$  is a sheaf, arrows already satisfy sheaf descent; this is the meaning of *prestack* used here. Object descent is not yet required.

*Stackification.* A stackification of a prestack is a stack receiving a morphism from it, universal for morphisms to stacks, up to the usual natural isomorphisms. For the action groupoid this construction is the definition of the quotient stack; see Definition 78.20.1, Tag 044Q. Informally it adds objects represented by compatible local objects, and identifies descriptions related by refinement.

*Torsors.* The unique object of  $(\mathcal{B}_0)_S$  corresponds to the trivial torsor  $G_S$ . This means that the one object is represented geometrically by the trivial family, and its arrows become symmetries of that family.

*Convention check.* The next calculation only keeps track of multiplication order; for the running example  $\mathbb{Z}/2\mathbb{Z}$ , the group is commutative and every element equals its inverse, so this distinction disappears. A left-equivariant automorphism of  $G_S$  has the form  $R_h(x) = xh$ . Since

$$R_k \circ R_h = R_{hk},$$

$\mathrm{Aut}_{G_S}(G_S)$  is naturally  $G(S)^{\mathrm{op}}$  via  $h \mapsto R_h$ . The functor from the one-object groupoid with composition  $g_2 \circ g_1 = g_2 g_1$  sends  $g$  to  $R_{g^{-1}}$ ; indeed

$$R_{g_2^{-1}} \circ R_{g_1^{-1}} = R_{(g_2 g_1)^{-1}}.$$

This makes the left/right convention explicit.

Every torsor is locally a trivial one, and its local identifications are the transition automorphisms of those trivial objects. Once the stack axioms are proved below, this identifies the resulting stack with the stackification of  $\mathcal{B}_0$ . The canonical equivalence is Lemma 95.15.4, Tag 0CQJ; it is a comparison with the notation, not an input into our proof of descent.

For  $G = \mathbb{Z}/2\mathbb{Z}$ , the naive fiber contains just a labeled pair of sheets, while stackification admits all locally split double covers. Section 18 gives explicit nonsplit examples.

**Exercise 5** (Opposite-group check). For a possibly noncommutative  $G$ , verify the two displayed composition formulas. Explain why  $g \mapsto R_g$  is not generally a functor from the one-object groupoid with its usual composition, and why  $g \mapsto R_{g^{-1}}$  is.

### Checkpoint.

- ◇ Distinguished: the one-object groupoid, the naive prestack, its stackification, and the torsor description.
- ◇ Check: explain why a nonsplit double cover is missing from the naive fiber.

## 9 Checking that BG is an ordinary category

### 9.1 Objects, arrows, and equivariance

We return to Definition 1. Its objects are the pairs  $(S, P)$ , with  $P \rightarrow S$  a  $G_S$ -torsor. We now understand all the ingredients. We spell out equivariance for its arrows, then supply identities and composition and verify the category axioms.

For  $f : S \rightarrow T$ , write

$$\alpha_f = \text{id}_G \times f : G_S \rightarrow G_T.$$

A prospective arrow  $(S, P) \rightarrow (T, Q)$  is a pair  $(f, \tilde{f})$  with  $f$  a  $B$ -morphism and

$\tilde{f} : P \rightarrow Q$  fitting into

$$\begin{array}{ccc} P & \xrightarrow{\tilde{f}} & Q \\ \pi_P \downarrow & & \downarrow \pi_Q \\ S & \xrightarrow{f} & T \end{array}$$

The required equivariance means that the following diagram commutes:

$$\begin{array}{ccc} G_S \times_S P & \xrightarrow{a_P} & P \\ \alpha_f \times \tilde{f} \downarrow & & \downarrow \tilde{f} \\ G_T \times_T Q & \xrightarrow{a_Q} & Q \end{array}$$

In test-point notation this is  $\tilde{f}(g \cdot p) = \alpha_f(g) \cdot \tilde{f}(p)$ . The factor  $\alpha_f$  is necessary because the groups act over different bases.

### 9.2 Category axioms

Define

$$\text{id}_{(S,P)} = (\text{id}_S, \text{id}_P).$$

Both required diagrams commute for this pair. For successive arrows  $(f, \tilde{f})$  and  $(g, \tilde{g})$ , define

$$(g, \tilde{g})(f, \tilde{f}) = (gf, \tilde{g}\tilde{f}).$$

The composite lies over  $gf$  since

$$\pi_R \tilde{g}\tilde{f} = g\pi_Q \tilde{f} = gf\pi_P.$$

Its equivariance follows from  $\alpha_{gf} = \alpha_g \alpha_f$ :

$$\begin{aligned} \tilde{g}\tilde{f}(h \cdot p) &= \tilde{g}(\alpha_f(h) \cdot \tilde{f}(p)) \\ &= \alpha_{gf}(h) \cdot \tilde{g}\tilde{f}(p). \end{aligned}$$

For three arrows, the base components of the two parenthesizations agree by associativity in  $\text{Sch}/B$ , and the total-space components agree by associativity of scheme morphisms. Thus composition is associative. Composing on the left with  $(\text{id}_T, \text{id}_Q)$ , or on the right with  $(\text{id}_S, \text{id}_P)$ , leaves both components of  $(f, \tilde{f})$  unchanged. This proves both identity laws.

### Checkpoint.

- ◇ Defined: objects, arrows, identities, and composition.
- ◇ Verified: equivariance of composites, associativity, and both identity laws.
- ◇ Result:  $BG$  is an ordinary category.

For double covers, equivariance says  $\tilde{f}\sigma_P = \sigma_Q\tilde{f}$ . Identities and composites commute with the involutions by the equations just checked. We do not define arrows to be arbitrary maps of degree-two covers with the actions forgotten.

**Example 14** (Moving the base and exchanging the sheets). Let  $B = \text{Spec } k$ ,  $S = \mathbb{A}_k^1$ , and  $P = S_0 \amalg S_1$  with its sheet-swap action. The following square defines a total-category arrow  $(S, P) \rightarrow (S, P)$ :

$$\begin{array}{ccc} S_0 \amalg S_1 & \xrightarrow{\tilde{f}} & S_0 \amalg S_1 \\ \pi \downarrow & & \downarrow \pi \\ S & \xrightarrow{f} & S. \end{array}$$

Here  $f$  is the squaring map, and on labeled test points

$$f(t) = t^2, \quad \tilde{f}(t, a) = (t^2, a + 1).$$

Each component map is the scheme morphism defined by  $k[t] \rightarrow k[t]$ ,  $t \mapsto t^2$ . The equation expressing equivariance is

$$\tilde{f}(t, a + 1) = (t^2, a) = \sigma \tilde{f}(t, a).$$

This arrow is not invertible: an inverse pair would require an inverse to  $f$ , but the image of its ring map does not contain  $t$ . In contrast,  $\sigma(t, a) = (t, a + 1)$  together with  $\text{id}_S$  defines an invertible arrow whose square is the identity. Composing the displayed arrow with itself gives  $(t, a) \mapsto (t^4, a)$ : the two sheet exchanges cancel, but the composite base map is not the identity. The two components of a total-category arrow carry different information.

**Exercise 6** (Category axioms). Write the complete verification of the identity laws and associativity for three pairs  $(f, \tilde{f})$ . Explain why defining arrows as actual maps  $P \rightarrow Q$  avoids having to insert associativity isomorphisms of fiber products in the definition of composition.

**Exercise 7** (All arrows between trivial torsors). Fix  $f : S \rightarrow T$ . Show that an equivariant map  $G_S \rightarrow G_T$  over  $f$  has the form

$$(x, s) \mapsto (xh(s), f(s)), \quad h \in G(S).$$

If a second arrow over  $g : T \rightarrow U$  is determined by  $k \in G(T)$ , compute the section determining the composite.

## 10 The projection to schemes

The definition of a category over a base now applies to our constructed category. The data already include base objects and base arrows; the remaining obligation is to check functoriality. Define

$$p : BG \longrightarrow \text{Sch}/B, \quad p(S, P) = S,$$

and  $p(f, \tilde{f}) = f$ . We have

$$p(\text{id}_{(S,P)}) = \text{id}_S$$

and

$$\begin{aligned} p((g, \tilde{g})(f, \tilde{f})) &= gf \\ &= p(g, \tilde{g})p(f, \tilde{f}). \end{aligned}$$

Thus this is a functor. It forgets the torsor but retains the base map of each arrow. The terminology “over  $S$ ” and “over  $f$ ” now refers precisely to this functor.

### Checkpoint.

- ◇ Defined:  $p(S, P) = S$  and  $p(f, \tilde{f}) = f$ .
- ◇ Verified: preservation of identities and composition.
- ◇ Result:  $p : BG \rightarrow \text{Sch}/B$  is a functor.

For a double cover,  $p$  remembers  $S$  and forgets both sheets and the involution. In particular, forgetting the involution is not the same operation as taking isomorphism classes of covers.

**Exercise 8** (An arrow over a nonidentity endomorphism). Assume  $B \neq \emptyset$ . Take  $S = \mathbb{A}_B^1$  with its endomorphism  $f$  given by  $t \mapsto t^2$ . Use trivial torsors to construct an arrow  $(S, G_S) \rightarrow (S, G_S)$  over  $f$ . Why is it not a morphism in the fiber  $(BG)_S$ ?

## 11 Fiber categories and pull-back torsors

### 11.1 The fiber, without yet asserting invertibility

By definition,  $(BG)_S$  has  $G_S$ -torsors as objects and equivariant  $S$ -maps as arrows. In other words, the arrows have base component  $\text{id}_S$ . At this point we do not build invertibility into their definition. It will be proved in Section 13.

### 11.2 Keep the types of objects separate

The following similar-looking symbols describe different kinds of mathematical objects:

- ◇  $G_S$  is a *scheme over  $S$* , with a group law.
- ◇  $(BG)_S$  is a *category*: its objects are torsors over  $S$ , and its arrows are equivariant  $S$ -maps between them.
- ◇  $G_S(T) = \text{Hom}_S(T, G_S)$  is a *set with a group law*, once a particular morphism  $T \rightarrow S$  has been specified.
- ◇  $P_T = P \times_S T$  is a *scheme over  $T$* , obtained from a particular torsor  $P \rightarrow S$ ; it is one object of  $(BG)_T$ .

For the trivial double cover,  $G_S$  and  $P$  can be the same underlying  $S$ -scheme. Neither is the category  $(BG)_S$ : that category also includes other torsors over  $S$  and the maps between them. The subscript on  $(BG)_S$  means “keep the base fixed at  $S$ ,” whereas the parentheses in  $G_S(T)$  mean “take maps from  $T$ .”

### 11.3 Base change of the geometric data

Given  $f : T \rightarrow S$  and a torsor  $P \rightarrow S$ , we must produce an object over  $T$ . Set

$$f^*P = P \times_S T, \quad G_T = G_S \times_S T.$$

The base change of the action is

$$g \cdot (p, t) = (\alpha_f(g) \cdot p, t).$$

The unit and associativity equations hold because their diagrams are base changes of the action diagrams on  $P$ . If  $\{U_a \rightarrow S\}$  trivializes  $P$ , then  $\{U_a \times_S T \rightarrow T\}$  is an fppf cover, and on it  $f^*P$  is the regular  $G_T$ -torsor. Consequently  $f^*P \rightarrow T$  is a torsor. The morphism  $\Theta_{f^*P}$  is also the base change of  $\Theta_P$ , so it is an isomorphism.

Projection gives an equivariant arrow

$$(f, \text{pr}_P) : (T, f^*P) \longrightarrow (S, P). \quad (11.1)$$

We have constructed a lift of  $f$ ; it remains to prove its universal property in the total category.

#### Checkpoint.

- ◊ Constructed:  $f^*P$ , its action, and its projection to  $P$ .

- ◊ Verified: local triviality and equivariance.
- ◊ Still to check: the cartesian universal property.

For  $\mathbb{Z}/2\mathbb{Z}$ , pullback gives  $P \times_S T \rightarrow T$  with involution  $(p, t) \mapsto (\sigma_P(p), t)$ . Pulling back  $S \amalg S \rightarrow S$  yields  $T \amalg T \rightarrow T$ .

**Exercise 9** (A pullback computation). Let  $k$  have characteristic different from 2, and consider  $P \rightarrow S = \text{Spec } k[t, t^{-1}]$  given by  $u^2 = t$ . Pull it back by  $T = \text{Spec } k[v, v^{-1}] \rightarrow S, t = v^2$ . Write its coordinate algebra and exhibit its two sheets and involution.

## 12 Cartesian morphisms and the fibration

The pullback torsor has already been constructed. Its extra property is this: a map into the original torsor can be viewed as a map into the pulled-back torsor whenever its base map has been specified through the new base. There is exactly one way to do so. The word *cartesian* packages this existence-and-uniqueness assertion; it does not mean that every object mapping into the pullback is isomorphic to it.

**Definition 13.** An arrow  $u : x \rightarrow y$  in a category over  $\mathcal{C}$ , lying over  $f : T \rightarrow S$ , is *cartesian* if the following holds. Given  $z$  over  $V$ , an arrow  $v : z \rightarrow y$  over  $h : V \rightarrow S$ , and  $k : V \rightarrow T$  with  $h = fk$ ,

there is a unique arrow  $w : z \rightarrow x$  over  $k$  such that  $uw = v$ . A *fibred category* is a category over  $\mathcal{C}$  admitting a cartesian lift of every  $f : T \rightarrow S$  with every prescribed target over  $S$ .

The Stacks Project uses the term *strongly cartesian* for this property, which allows any factorization  $h = fk$  (Definition 4.33.1, Tag 02XK). Its fibration definition requires these lifts (Definition 4.33.5, Tag 02XM).

The data already constructed in (11.1) have the right source, target, and base map. We now verify the quantified factorization condition. Let  $(V, R)$  be any object, let  $(h, \tilde{h}) : (V, R) \rightarrow (S, P)$  be any arrow, and suppose  $h = fk$ . The competing diagrams are

$$\begin{array}{ccc} R & \xrightarrow{\tilde{h}} & P \\ & \searrow \lambda & \nearrow \text{pr}_P \\ & P \times_S T & \end{array}$$

over

$$\begin{array}{ccc} V & \xrightarrow{h} & S \\ & \searrow k & \nearrow f \\ & T & \end{array}$$

The two maps  $\tilde{h} : R \rightarrow P$  and  $k\pi_R : R \rightarrow T$  satisfy

$$\pi_P \tilde{h} = h\pi_R = fk\pi_R.$$

The universal property of  $P \times_S T$ , stated in Section 3, therefore gives a unique map

$$\begin{aligned} \lambda : R &\longrightarrow P \times_S T, \\ r &\longmapsto (\tilde{h}(r), k\pi_R(r)). \end{aligned}$$

It lies over  $k$ , and  $\text{pr}_P \lambda = \tilde{h}$ . For  $g \in G_V$ ,

$$\begin{aligned} \lambda(g \cdot r) &= (\alpha_h(g) \cdot \tilde{h}(r), k\pi_R(r)) \\ &= \alpha_k(g) \cdot \lambda(r), \end{aligned}$$

because  $\alpha_h = \alpha_f \alpha_k$ . Thus  $(k, \lambda)$  is an arrow of  $BG$ . Any other factorization must have the same projections to  $P$  and  $T$ , so the fiber-product uniqueness clause makes it equal to  $(k, \lambda)$ . This proves cartesianess.

In the double-cover picture, the map  $\lambda$  records two things: which point of the old cover  $P$  a point of  $R$  maps to, and which point of the new base  $T$  it is meant to lie above. The equation  $h = fk$  says these two records describe the same point of  $S$ . It makes the pair a point of the fiber product. The construction gives a map  $R \rightarrow P \times_S T$ , not an assertion that  $R$  equals that fiber product.

## 12.1 Unique pullbacks and their coherence

Suppose  $u : x \rightarrow y$  and  $u' : x' \rightarrow y$  are cartesian lifts of the same  $f : T \rightarrow S$ . Applying their universal properties gives vertical arrows  $a : x \rightarrow x'$  and  $b : x' \rightarrow x$  with  $u'a = u$  and  $ub = u'$ . Both  $ba$  and  $\text{id}_x$  are factorizations of  $u$  through  $u$  over  $\text{id}_T$ , so  $ba = \text{id}_x$ ; likewise  $ab = \text{id}_{x'}$ . The isomorphism  $a$  is unique *subject to compatibility with  $u, u'$* . This does not forbid other automorphisms of the underlying torsor.

For  $T \xrightarrow{f} S \xrightarrow{g} U$  and a torsor  $P \rightarrow U$ , the canonical comparison is

$$P \times_U T \xrightarrow{\sim} (P \times_U S) \times_S T,$$

$$(p, t) \mapsto ((p, f(t)), t).$$

Thus  $(gf)^*P \simeq f^*(g^*P)$ . Projection also gives  $(\text{id}_S)^*P \simeq P$ . The comparison maps are equivariant and respect the maps to  $P$ . For three successive base changes, all ways of inserting these comparisons agree: they have the same projections to each factor, so uniqueness in the fiber-product property proves coherence. The general categorical version is Lemma 4.33.7, Tag 02X0.

#### Checkpoint.

- ◊ Verified: a competing arrow factors uniquely through the pullback arrow over the prescribed base map.
- ◊ Result:  $BG \rightarrow \text{Sch}/B$  is fibered.
- ◊ Check: write both projections of the factorization.

For double covers, the factorization sends  $r$  to its image in the original cover together with its prescribed point of the new base. Successive base changes retain the same two sheets and give canonically isomorphic double covers.

**Exercise 10** (Complete the cartesian diagram). In the competing diagram above, prescribe  $R, \tilde{h}$ , and  $k$ . Write both projections of  $\lambda$ . Prove directly that equivariance uses  $\alpha_h = \alpha_f \alpha_k$ , and that no other

arrow over  $k$  can factor  $(h, \tilde{h})$  through (11.1).

**Exercise 11** (Uniqueness with a qualification). A trivial double cover has a non-trivial deck automorphism. Explain why this does not contradict uniqueness up to unique isomorphism of a cartesian lift. Which compatibility equation does the deck automorphism usually fail?

### 13 Why the fibers are groupoids

**Definition 14.** A groupoid is a category in which every arrow has an inverse. A *category fibered in groupoids* is a fibered category whose fibers are groupoids.

We already have a fibered category, and  $(BG)_S$  was defined using all equivariant  $S$ -maps. The remaining task is to prove their invertibility.

**Lemma 1** (Equivariant maps of torsors). *If  $P$  and  $Q$  are  $G_S$ -torsors, every equivariant  $S$ -map  $u : P \rightarrow Q$  is an isomorphism, whose inverse is equivariant.*

*Proof.* Take trivializing covers for both torsors and form their fiber products; this gives a common fppf trivializing cover  $\{U_a \rightarrow S\}$ . On each  $U_a$ , the map becomes an equivariant map  $G_{U_a} \rightarrow G_{U_a}$ . If  $h = u(e)$ , equivariance gives  $u(x) = xh$ . The map  $x \mapsto xh^{-1}$  is its inverse. Consequently  $u$  is an isomorphism after the cover and therefore globally by Tag 02L4. If  $q = u(p)$ , the equation  $u(gp) = gq$  im-

plies  $u^{-1}(gq) = gu^{-1}(q)$ , proving equivariance of the inverse.  $\square$

### Checkpoint.

- ◇ Proved: every equivariant map of torsors over the same base is invertible.
- ◇ Result: every fiber  $(BG)_S$  is a groupoid.
- ◇ Together with cartesian lifts:  $BG \rightarrow \text{Sch}/B$  is a category fibered in groupoids.

### 13.1 The equivalent description of total arrows

We can now justify an often-used alternative definition. Given  $(f, \tilde{f}) : (S, P) \rightarrow (T, Q)$ , the commutative square induces

$$\begin{aligned} \hat{f} : P &\longrightarrow f^*Q = Q \times_T S, \\ p &\longmapsto (\tilde{f}(p), \pi_P(p)). \end{aligned}$$

It is an equivariant  $S$ -map between torsors, hence an isomorphism by Lemma 1. Conversely, such an isomorphism followed by projection to  $Q$  yields  $\tilde{f}$ . Therefore total arrows are equivalently base maps  $f$  equipped with  $P \simeq f^*Q$ ; see also Examples of Stacks, Section 95.14.8, Tag 04UR.

Every total arrow is consequently cartesian: it is the composite of a vertical isomorphism with a cartesian pull-back arrow. This also verifies the alternative convention for a category fibered

in groupoids in which all arrows are required to be cartesian. Notice that  $P \rightarrow Q$  itself need not be an isomorphism when the base map is not an isomorphism.

For the constant group of order two, an arrow in the fiber is locally  $x \mapsto x + \varepsilon$  with  $\varepsilon \in \mathbb{Z}/2\mathbb{Z}$ . It is either the identity or the sheet swap on each connected component. This is the concrete form of the lemma.

**Exercise 12** (Invertibility without defining it). Prove Lemma 1, including the construction of a common trivializing cover. Why is checking bijectivity only on geometric fibers not a substitute for the scheme-theoretic proof? Give a total arrow whose map on total spaces is not an isomorphism.

### 14 The definition of a stack

The available structure is now a category fibered in groupoids. A choice of cartesian lifts is called a *cleavage*; it makes notation such as  $f^*x$  convenient. Its canonical comparison isomorphisms will always be inserted when comparing iterated pull-backs.

**Definition 15** (The two stack conditions). A category fibered in groupoids  $\mathcal{F}$  on our site is a stack if:

- (1) Isomorphisms form a sheaf. For  $x, y \in \mathcal{F}_S$ , define

$$T \longmapsto \text{Isom}_{\mathcal{F}_T}(x_T, y_T).$$

This presheaf must be a sheaf for fppf covers of  $S$ -schemes.

- (2) For every fppf cover  $\{S_i \rightarrow S\}$ , every descent datum for objects is effective.

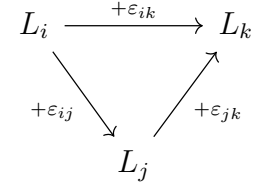
This is Definition 8.5.1, Tag 02ZI, with its fibration condition already established. We spell out the terminology in the second clause.

#### 14.1 First picture: an ordinary open cover

For the moment let the  $S_i$  be open subsets covering  $S$ . Suppose a double cover is given over each  $S_i$ . On  $S_i \cap S_j$ , an isomorphism says which sheet in the first description corresponds to which sheet in the second. On a triple intersection, changing from the first description to the third directly must have the same result as changing first to the second and then to the third. This is the *cocycle condition*: the identifications must not depend on the route taken.

To *descend* this information means to construct a single double cover over  $S$  with these local descriptions. The datum is called *effective* when such a global cover actually exists. For general fppf covers, the same questions make sense, but  $S_i \times_S S_j$  takes the place of the intersection. Its two projections keep track of which local description is being used. These projections must be retained because the covering maps need not be inclusions.

**Example 15** (Two routes through the sheet labels). Suppose the local double covers have been labeled on the open sets  $S_i$ . On a triple overlap  $V = S_i \cap S_j \cap S_k$ , each description is  $V \amalg V$ ; call the three labeled descriptions  $L_i, L_j, L_k$ . A transition from labels  $i$  to labels  $j$  adds a locally constant value  $\varepsilon_{ij}$  in  $\mathbb{Z}/2\mathbb{Z}$ . The two proposed routes from  $L_i$  to  $L_k$  are



Starting with label  $a$ , the lower route gives  $a + \varepsilon_{ij} + \varepsilon_{jk}$ , and the direct route gives  $a + \varepsilon_{ik}$ . The diagram commutes precisely when

$$\varepsilon_{ij} + \varepsilon_{jk} = \varepsilon_{ik} \quad \text{on } V.$$

For instance, if the first two transitions swap sheets, the direct transition must preserve them. Choosing all three transitions to be swaps on a nonempty  $V$  is inconsistent: the lower route sends 0 to 0, but the direct route sends 0 to 1. Each proposed map individually is a valid isomorphism; it is their joint compatibility that fails. This is the extra information recorded by a cocycle.

#### 14.2 The general datum

Put  $S_{ij} = S_i \times_S S_j$  with projections  $r_1, r_2$ , and put  $S_{ijk} = S_i \times_S S_j \times_S S_k$ . On triple

overlaps let  $q_{ab}$  select factors  $a, b$ . A *descent datum* consists of  $x_i \in \mathcal{F}_{S_i}$  and isomorphisms

$$\varphi_{ij} : r_1^* x_i \xrightarrow{\sim} r_2^* x_j$$

such that

$$q_{23}^* \varphi_{jk} \circ q_{12}^* \varphi_{ij} = q_{13}^* \varphi_{ik}. \quad (14.1)$$

All arrows in this equation are over  $S_{ijk}$ . This is Definition 8.3.1, Tag 026B, with projection indices starting at 1.

The datum is *effective* if there is  $x \in \mathcal{F}_S$  with  $\psi_i : x|_{S_i} \simeq x_i$  for which

$$\varphi_{ij} = (r_2^* \psi_j) \circ (r_1^* \psi_i)^{-1}. \quad (14.2)$$

The two pullbacks of  $x$  in the middle are identified canonically.

### 14.3 Self-overlaps require care

An fppf cover need not consist of open immersions. Thus  $S_{ii}$  need not equal  $S_i$ , and the two objects  $r_1^* x_i, r_2^* x_i$  on  $S_{ii}$  are different pullbacks. The correct normalization is

$$\Delta_i^* \varphi_{ii} = \text{id}_{x_i}, \quad (14.3)$$

where  $\Delta_i : S_i \rightarrow S_i \times_S S_i$ . If  $s_{ij} : S_{ij} \rightarrow S_{ji}$  exchanges factors, then

$$s_{ij}^* \varphi_{ji} = \varphi_{ij}^{-1}.$$

To prove (14.3), pull the  $(i, i, i)$  cocycle back along the small diagonal. The resulting invertible endomorphism  $a$  satisfies  $a^2 = a$ , hence  $a = \text{id}$ . Pulling the  $(i, j, i)$  cocycle back along  $(u, v) \mapsto (u, v, u)$  then proves the inverse formula. These are Remarks 8.3.2, Tag 026C. The assertion “ $\varphi_{ii} = \text{id}$  on all of  $S_{ii}$ ” is in general incorrect.

### 14.4 The categorical meaning of descent

Morphisms between descent data are families  $u_i : x_i \rightarrow y_i$  such that

$$\eta_{ij} \circ r_1^* u_i = r_2^* u_j \circ \varphi_{ij},$$

where  $\eta_{ij}$  is the transition system for  $y_i$ . They form the descent groupoid  $\text{Desc}_{\mathcal{F}}(\{S_i \rightarrow S\})$ . Restriction is a functor

$$\mathcal{F}_S \longrightarrow \text{Desc}_{\mathcal{F}}(\{S_i \rightarrow S\}).$$

The Isom-sheaf axiom says it is fully faithful: compatible local arrows between two existing global objects come from unique global arrows. Effective descent says it is essentially surjective: each local object system has a global realization. Both are required for an equivalence.

#### Checkpoint.

- ◇ Defined: the Isom-sheaf axiom, descent datum, cocycle, and effective descent.
- ◇ Check: state separately what must glue in each stack axiom.
- ◇ Still to prove: both gluing assertions for  $BG$ .

**Exercise 13** (A fibered groupoid with insufficient arrow descent). On the open-set site of a space consisting of two discrete points, set  $A(U) = \mathbb{Z}/2\mathbb{Z}$  for

nonempty  $U$  and  $A(\emptyset) = \{0\}$ , with the evident restrictions. Form the one-object groupoid with automorphism group  $A(U)$  over every open  $U$ . Explain why its associated total category is fibered in groupoids but its automorphisms do not form a sheaf.

## 15 Isomorphisms form a sheaf

### 15.1 The presheaf and its exact restriction laws

The objects  $P$  and  $Q$  are already defined over all of  $S$ . What will vary locally is an *identification between them*. Over a  $B$ -scheme  $T \rightarrow S$ , such an identification is an equivariant isomorphism  $P_T \rightarrow Q_T$ . Thus the elements of the set we are about to define are maps, not points of either cover.

Restriction has a concrete meaning. Given  $u : T' \rightarrow T$ , pull back the map  $\varphi : P_T \rightarrow Q_T$  along  $u$ . On the corresponding fiber products it sends  $(p, t')$  to  $(\varphi(p), t')$ : the identification of the covers is unchanged, but the base has been changed to  $T'$ . The comparison maps below only identify these iterated fiber products with the previously chosen notation  $P_{T'}$  and  $Q_{T'}$ .

Fix  $G_S$ -torsors  $P, Q$ . Define

$$\begin{aligned} \mathcal{I}(T) &= \underline{\text{Isom}}_G(P, Q)(T) \\ &= \text{Isom}_{G_T}(P_T, Q_T). \end{aligned}$$

For  $u : T' \rightarrow T$  over  $S$ , let

$$c_{P,u} : P_{T'} \xrightarrow{\sim} u^*(P_T)$$

be the canonical comparison, and similarly for  $Q$ . Define the restriction function by

$$\rho_u(\varphi) = c_{Q,u}^{-1} \circ u^* \varphi \circ c_{P,u}.$$

It is a map between the chosen objects  $P_{T'}$  and  $Q_{T'}$ . Identity comparisons give  $\rho_{\text{id}} = \text{id}$ . If  $v : T'' \rightarrow T'$ , coherence of the fiber-product comparisons gives  $\rho_v \rho_u = \rho_{uv}$  as an *equality of functions*. Thus  $\mathcal{I}$  is an actual set-valued presheaf, not a presheaf only up to isomorphism. This construction is formalized in Lemma 8.2.1 and Definition 8.2.2, Tag 026A and Tag 02ZB.

### 15.2 The external gluing theorem

**Theorem 9** (Descent for scheme morphisms). *Let  $X, Y$  be schemes over a scheme  $S$ , and let  $\{S_i \rightarrow S\}$  be an fpqc cover. Compatible  $S_i$ -maps  $X_{S_i} \rightarrow Y_{S_i}$  glue to a unique  $S$ -map  $X \rightarrow Y$ . In particular, equality of maps can be checked after the cover. No affineness assumption on  $X$  or  $Y$  is required.*

This is Lemma 35.35.11 and its Hom-sheaf formulation in Remark 35.35.12, Tag 02W0 and Tag 040L. One underlying reason is that representable functors are fpqc sheaves:  $X_{S_i} \rightarrow X$  is a cover, and compatible maps from these schemes to  $Y$  glue by the sheaf axiom for  $h_Y$ . Compatibility with the maps to  $S$  is checked by the same uniqueness statement.

### 15.3 Apply the theorem to torsor isomorphisms

The available data are an fppf cover  $\{S_i \rightarrow S\}$  and equivariant isomorphisms  $\varphi_i : P_{S_i} \simeq Q_{S_i}$  agreeing on overlaps. We must glue them and verify equivariance, invertibility, and uniqueness.

Theorem 9 applies because  $P, Q$  are schemes and the cover is fpqc. It produces a unique underlying  $S$ -morphism  $\varphi : P \rightarrow Q$ . To check equivariance, compare

$$\varphi a_P, \quad a_Q(\mathrm{id}_G \times \varphi)$$

as maps  $G_S \times_S P \rightarrow Q$ . Their base changes agree because every  $\varphi_i$  is equivariant. The equality part of the theorem makes them equal globally.

The local inverses  $\varphi_i^{-1}$  agree on overlaps and glue to  $\chi : Q \rightarrow P$ . The maps  $\chi\varphi$  and  $\mathrm{id}_P$  agree on the cover, as do  $\varphi\chi$  and  $\mathrm{id}_Q$ . Hence they are equal globally and  $\varphi$  is an isomorphism. Its inverse is equivariant by the same equation or by Lemma 1. Finally, uniqueness of the underlying map gives uniqueness among equivariant isomorphisms.

To prove the sheaf axiom on the *whole* slice site, repeat this argument over every  $T \rightarrow S$  and every fppf cover of  $T$ , with  $P_T, Q_T$  in place of  $P, Q$ .

#### Checkpoint.

- ◇ Constructed: the unique glued map and its inverse.
- ◇ Verified: equivariance and both inverse equations.

- ◇ Result:  $\underline{\mathrm{Isom}}_G(P, Q)$  is an fppf sheaf, the first stack axiom.

For trivial double covers, a section of this sheaf is a locally constant choice to preserve or exchange the two labels. Compatible choices on a cover glue uniquely. For two nontrivial double covers, the same argument glues their local maps commuting with the involutions; no global labeling of either cover is required.

**Exercise 14** (The presheaf law). Using the comparisons  $c_{P,u}$ , verify  $\rho_v \rho_u = \rho_{uv}$ . Explain why the restriction law is an equality even though iterated fiber products are only canonically isomorphic.

**Exercise 15** (Gluing). Write the two maps whose equality expresses equivariance of a glued  $P \rightarrow Q$ . Then prove invertibility by gluing local inverses. Identify which clauses of Theorem 9 are used for existence and for uniqueness.

## 16 Effective descent for torsors

### 16.1 The affine descent input

Here *affine over  $S$*  is a property of the morphism to  $S$ ; it does not require the total space to be an affine scheme. For example,  $S \amalg S \rightarrow S$  is affine: above an affine open  $V = \mathrm{Spec} A$  it is  $V \amalg V = \mathrm{Spec}(A \times A)$ . This remains true when  $S = \mathbb{P}_k^1$ , although  $S \amalg S$  is not an affine scheme. The affine descent theorem concerns this relative meaning of affine.

The remaining stack condition starts with local objects, so it must construct a new global scheme. We use the following theorem as a black box.

**Theorem 10** (Effective descent of affine morphisms). *Let  $\{S_i \rightarrow S\}$  be an fpqc cover. Suppose  $X_i \rightarrow S_i$  are affine morphisms with isomorphisms  $r_1^* X_i \simeq r_2^* X_j$  satisfying the cocycle condition. There exists an affine  $S$ -scheme  $X$  and isomorphisms  $X_{S_i} \simeq X_i$  recovering those transitions. Together with Theorem 9, this gives an equivalence between affine  $S$ -schemes and their descent data.*

This is Lemma 35.37.1, Tag 0245. The theorem depends on faithfully flat descent of modules and algebras, not on torsor stackness. In affine coordinates it descends an algebra with compatible identifications to an algebra over the base ring; taking its spectrum constructs the global affine scheme. A cocycle is needed so that these identifications are mutually consistent.

We also use Lemma 35.23.20, Tag 02L5: if a morphism of schemes becomes affine over an fpqc cover of its target, it was affine already. This assertion concerns a *given* global morphism; it is distinct from the effective-descent theorem, which constructs a global object.

## 16.2 The datum and the underlying scheme

Suppose given  $G_{S_i}$ -torsors  $P_i \rightarrow S_i$  and equivariant  $\varphi_{ij} : r_1^* P_i \simeq r_2^* P_j$  satisfying (14.1). They prescribe how local torsors

should be identified on repeated base changes. We seek  $(P, \psi_i)$  satisfying (14.2) and all torsor axioms.

Each  $P_i \rightarrow S_i$  is affine: a trivializing fppf cover makes it isomorphic to the affine morphism  $G_{S_i} \rightarrow S_i$ , and Tag 02L5 applies. The outer cover is fpqc by Tag 022C. Thus every hypothesis of Theorem 10 is verified. It gives an affine  $P \rightarrow S$  and isomorphisms

$$\psi_i : P_{S_i} \xrightarrow{\sim} P_i$$

recovering the specified transitions. At this stage  $P$  is a scheme over  $S$ ; its action is still to be constructed.

## 16.3 Descend the action and its equations

Transport each action  $a_i : G_{S_i} \times P_i \rightarrow P_i$  to

$$a'_i = \psi_i^{-1} a_i (\text{id}_G \times \psi_i)$$

on  $(G_S \times_S P)_{S_i} \rightarrow P_{S_i}$ . On  $S_{ij}$ , equivariance of  $\varphi_{ij}$  says

$$\varphi_{ij} a_i = a_j (\text{id}_G \times \varphi_{ij}).$$

Substitution of (14.2) shows that  $a'_i$  and  $a'_j$  agree. Theorem 9, applied with  $X = G_S \times_S P$  and  $Y = P$ , produces a unique map  $a : G_S \times_S P \rightarrow P$ .

Associativity compares the two maps from  $G_S \times_S G_S \times_S P$  to  $P$ :

$$a(m \times \text{id}_P), \quad a(\text{id}_G \times a).$$

They agree after every  $S_i$  because  $a_i$  is an action. Thus they agree globally. The

same equality test proves  $a(e \times \text{id}_P) = \text{id}_P$ . We have obtained a group action, and the  $\psi_i$  are equivariant by the way it was constructed.

#### 16.4 Local triviality and the torsor identity

For each  $i$ , choose an fppf cover  $\{U_{ia} \rightarrow S_i\}$  trivializing  $P_i$ . Composition of fppf coverings makes  $\{U_{ia} \rightarrow S\}_{i,a}$  a cover. On it,

$$P_{U_{ia}} \simeq (P_i)_{U_{ia}} \simeq G_{U_{ia}}$$

equivariantly. Consequently  $P \rightarrow S$  is locally trivial in the chosen topology. Proposition 3 also gives its surjectivity, flatness, and local finite presentation.

For an explicit check of the torsor identity, form

$$\Theta_P : G_S \times_S P \rightarrow P \times_S P.$$

Under  $\psi_i$  its base change is  $\Theta_{P_i}$ , hence an isomorphism. The isomorphism-locality theorem Tag 02L4 therefore makes  $\Theta_P$  an isomorphism. This verifies the requested identity globally, in addition to local triviality.

#### 16.5 Recovery and compatible uniqueness

The maps  $\psi_i$  already satisfy (14.2), by the affine descent construction, and they are now maps of torsors. Suppose  $(P', \psi'_i)$  is another realization. Define

$$\lambda_i = (\psi'_i)^{-1} \psi_i : P_{S_i} \simeq P'_{S_i}.$$

The two recovery equations imply agreement of  $\lambda_i$  and  $\lambda_j$  on overlaps. Section 15 gives a unique equivariant isomorphism  $\lambda : P \simeq P'$  such that

$$\psi'_i \lambda_{S_i} = \psi_i$$

for every  $i$ . The compatibility equations, not a claim that torsors have no automorphisms, are what force uniqueness.

#### 16.6 The sheaf interpretation

There is also a useful way to see the gluing before testing representability. On a test  $T \rightarrow S$ , put  $T_i = T \times_S S_i$  and consider tuples

$$(p_i) \in \prod_i P_i(T_i)$$

such that  $\varphi_{ij}(p_i) = p_j$  on  $T \times_S S_{ij}$ . These sets, with restriction of tuples, form a sheaf  $\mathcal{P}$ . Indeed, compatible tuples on a cover of  $T$  glue componentwise by the sheaf property of each  $P_i$ , and their compatibility equations can be checked on that cover. Over  $S_j$ , restriction to the diagonal of the  $j$ -th component gives  $\mathcal{P}|_{S_j} \simeq h_{P_j}$ ; the inverse transports that section to the other components by  $\varphi_{ji}$ . The cocycle proves compatibility and that these maps are inverse. The action is componentwise.

This argument constructs a sheaf torsor. Affine descent is the separate representability step, and identifies  $\mathcal{P}$  with  $h_P$ . Thus even this version uses no circular appeal to the stack of torsors.

**Checkpoint.**

- ◇ Constructed: the global scheme, action, and local identifications.
- ◇ Verified: action laws, local triviality, the torsor identity, and recovery of transitions.
- ◇ Result: effective descent, with uniqueness compatible with the local identifications.

For trivial double covers the transition maps are  $x \mapsto x + \varepsilon_{ij}$ ; their cocycle says the parity of sheet swaps is independent of the route through a triple overlap. Affine descent constructs the double cover, and equivariance glues its involution. This remains valid for a singleton fppf cover with a nontrivial self-overlap.

**Exercise 16** (A descent datum is more than local objects). Write a torsor descent datum using  $r_1, r_2$  and  $q_{12}, q_{23}, q_{13}$ . For trivial double covers interpret the cocycle as an equation for  $\varepsilon_{ij}$ . On a triple overlap, why is the assignment  $\varepsilon_{ij} = \varepsilon_{jk} = 1, \varepsilon_{ik} = 1$  inconsistent?

**Exercise 17** (The self-overlap error). Let  $U = P \rightarrow S$  be a double-cover torsor and use its tautological section to trivialize  $P_U$ . On  $U \times_S U$ , define  $\varepsilon(u, v)$  by  $v = \varepsilon(u, v) \cdot u$ . Show that this is its transition function, that it vanishes on the diagonal, and that it is 1 on the graph of the deck involution. Why does this refute  $\varphi_{11} = \text{id}$  on the entire self-overlap?

**Exercise 18** (Descend a structure map). Assume affine descent has produced  $P$

and  $\psi_i$ . Prove in detail that the transported actions  $a'_i$  agree, and list the domains of the two morphisms used to check associativity. Is checking the action only on geometric points sufficient?

**Exercise 19** (Uniqueness of a realization). For two realizations  $(P, \psi_i)$  and  $(P', \psi'_i)$ , prove that  $(\psi'_i)^{-1}\psi_i$  agrees on overlaps. State exactly the uniqueness assertion that follows, and explain why omitting the  $\psi_i$  changes it.

## 17 The stack theorem

We now apply Definition 15 to the structure we have built. The required data and their verifications are all available:

- (1) Section 9 established an ordinary category  $BG$ .
- (2) Section 10 established the functor to  $\text{Sch}/B$ .
- (3) Sections 11–12 constructed cartesian lifts of all base maps.
- (4) Section 13 proved invertibility in every fiber.
- (5) Section 15 proved the Isom-sheaf condition on every slice site.
- (6) Section 16 proved effective object descent.

These prove Theorem 8.

The functor from the naive prestack to this torsor category is fully faithful on

fibers: its arrows are exactly the translations described in Section 8. Every torsor is locally in its image. Since the target is now proved to be a stack, the stackification comparison of Tag 0CQJ identifies it with  $[B/G]$ .

**Checkpoint.**

- ◊ Verified: a category over  $\text{Sch}/B$ , cartesian lifts, and groupoid fibers.
- ◊ Verified: both descent axioms.
- ◊ Result:  $BG = [\text{pt}/G]$  is a stack for the fppf covering rule.

For the constant group of order two, the theorem says that equivariant isomorphisms of double covers glue uniquely and that every compatible system of local double covers has a global realization.

## 18 A detailed double-cover example

### 18.1 The groupoid of double covers

Let  $G = (\mathbb{Z}/2\mathbb{Z})_B$ . A torsor is finite *étale* of degree 2: it has this property on a trivializing cover, and finite locally free morphisms of fixed degree and *étale* morphisms descend fpqc-locally (Lemmas 35.23.32 and 35.23.31, Tag 02V0 and Tag 02VN). Conversely, a finite *étale* degree-two cover is *étale*-locally two copies of its base (Lemma 41.18.3, Tag 04HN). The unique permutation exchanging the two points commutes with every permutation of two points, so its

local involutions glue, independently of labeling, to a canonical action of  $\mathbb{Z}/2\mathbb{Z}$ . This is a torsor action.

For the trivial cover over  $S$ , an automorphism is  $x \mapsto x + \varepsilon$  with  $\varepsilon \in G(S)$ . If  $S$  is nonempty and connected, there are exactly two choices. On a disconnected base the choice can differ on each open-and-closed component; it is incorrect to count exactly two global automorphisms for every base.

### 18.2 A nonsplit cover and its splitting cover

Take a field  $k$  with  $\text{char } k \neq 2$ , let

$$\begin{aligned} S &= \text{Spec } k[t, t^{-1}], \\ P &= \text{Spec } k[u, u^{-1}], \end{aligned}$$

and use  $t = u^2$  for the structural map. The coordinate algebra is free of rank two over  $k[t, t^{-1}]$ , with basis  $1, u$ , and the derivative  $2u$  is a unit. The standard *étale* algebra criterion says that a monic polynomial with invertible derivative in its quotient algebra defines an *étale* algebra (Definition 10.144.1 and Lemma 10.144.2(1), Tag 00UB and Tag 00UC). Thus our map is finite *étale*. The action is  $u \mapsto -u$ .

The torsor identity can be checked without counting points: the algebra of  $P \times_S P$  is

$$k[u^{\pm 1}, v^{\pm 1}]/(u^2 - v^2).$$

The ideals  $(u - v)$  and  $(u + v)$  are comaximal because  $2v$  is invertible. Thus this

scheme is the disjoint union of the diagonal and the graph of the involution, exactly the two components of  $G_S \times_S P$ .

A section of  $P \rightarrow S$  would give  $w \in k[t, t^{-1}]$  with  $w^2 = t$ . Every unit has the form  $ct^n$ , so this would force  $2n = 1$ . There is no section, and hence the torsor is nontrivial by Exercise 3. Nevertheless it splits on the single fppf cover  $P \rightarrow S$ . Its self-overlap records whether  $v = u$  or  $v = -u$ . The transition  $\varepsilon(u, v)$  equals 0 or 1 respectively, and on a triple overlap

$$\varepsilon(u, v) + \varepsilon(v, w) = \varepsilon(u, w).$$

The equality expresses the composition of signs. Descent of the trivial torsor over  $P$ , with this transition, reconstructs the original nonsplit torsor over  $S$ .

### 18.3 Characteristic two

Over a field  $k$  of characteristic 2, the morphism

$$\mathrm{Spec} k[x] \longrightarrow \mathrm{Spec} k[t], \quad t = x^2 - x,$$

is free of degree two, with invertible derivative  $-1$ . Its involution is  $x \mapsto x + 1$ . On the self-product the difference  $y - x$  satisfies

$$(y - x)^2 - (y - x) = 0,$$

which splits into the comaximal factors  $y - x$  and  $y - x - 1$ . This verifies the constant-group torsor identity in characteristic 2. A polynomial section would solve  $r(t)^2 - r(t) = t$ ; a nonconstant  $r$  has

even degree on the left and a constant  $r$  gives a constant. Thus the torsor is nontrivial.

By comparison, over  $k[t, t^{-1}]$  the map  $t = u^2$  in characteristic 2 is a  $\mu_2$ -torsor: it is finite faithfully flat and the ratio of two lifts is an element of  $\mu_2$ . It is not *étale*, and the formula  $u \mapsto -u$  becomes the identity, so it does not define a constant  $\mathbb{Z}/2\mathbb{Z}$ -torsor action.

**Exercise 20** (Automorphisms on a disconnected base). Let  $S = S_1 \amalg S_2$ , with both  $S_i$  nonempty and connected. Determine the automorphism group of the trivial constant-group double cover over  $S$ . Explain its sheaf interpretation.

**Exercise 21** (Artin–Schreier check). Verify the degree, *étale* property, involution, torsor identity, and nontriviality of  $t = x^2 - x$  in characteristic 2. Locate precisely where the analogous formula  $u \mapsto -u$  fails for the constant group.

## 19 Comparison with families of curves

### 19.1 The actual moduli problem

The Stacks Project studies curves in Section 99.15, Tag 0D4Y. Its category Curves (Situation 99.15.1, Tag 0D4Z) has objects  $X \rightarrow S$  with  $S$  a scheme,  $X$  an algebraic space, and the morphism flat, proper, finitely presented, of relative dimension at most one. No smoothness, connectedness, reducedness, or fixed genus is im-

posed. Relative dimension is in the sense of Definition 67.33.2, Tag 06LR. Here is the distinction from the dimension of the total space: a family over a curve may have a two-dimensional total space, while each fiber is only one-dimensional. The relative-dimension condition concerns those fibers. More precisely, at a point of the family one measures the dimension there in the fiber over its image in the base; the bound must hold at every point. Thus “at most one” also allows zero-dimensional fibers, rather than requiring every fiber to be a smooth, connected curve. We may restrict the base schemes to  $B$ -schemes throughout.

An arrow is a *cartesian* square

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \xrightarrow{f} & S \end{array}$$

If all commutative squares were allowed, arrows in a fixed fiber could include noninvertible endomorphisms; the fibers would not be groupoids.

## 19.2 The formal categorical correspondence

| $BG$                       | Curves                    |
|----------------------------|---------------------------|
| Torsor $P \rightarrow S$   | Family $X \rightarrow S$  |
| Equivariant arrow over $f$ | Cartesian square over $f$ |
| $P \simeq f^*Q$            | $X' \simeq f^*X$          |
| Pullback torsor            | Base-changed family       |
| Equivariant isomorphism    | Isomorphism of families   |
| Affine descent             | Algebraic-space descent   |

Identities and composites are defined componentwise; cartesian squares paste, so the result is a category. Forgetting  $X$  gives a functor to schemes. Base change gives the cartesian lift, by exactly the same factorization proof as Section 12. Flatness, properness, finite presentation, and the relative-dimension bound are stable under arbitrary base change. For the first three properties see Lemmas 67.30.4, 67.40.3, and 67.28.3, respectively: Tag 03M0, Tag 04WP, and Tag 03XR. For the dimension bound use Lemma 67.34.3(1), Tag 04NS. Finally a cartesian square over  $\text{id}_S$  identifies  $X'$  with  $S \times_S X = X$ , so each fiber is a groupoid.

The base-change and locality facts used here are stated below. Thus the formal categorical argument is identical, while closure of the class of allowed objects is a geometric theorem.

### 19.3 Isomorphism descent for curves

For global  $X, Y \rightarrow S$ , consider

$$T \longmapsto \text{Isom}_T(X_T, Y_T).$$

The relevant black box is Lemma 95.7.2, Tag 04UA: compatible local morphisms between two algebraic spaces over a scheme glue uniquely on an fppf covering. It applies to these  $X, Y$  even if they are not schemes. Gluing inverses and checking the two identity composites on the cover proves the Isom-sheaf condition. No representability theorem for this Isom sheaf is needed for stackness.

### 19.4 Effective descent and the geometric conditions

The required object-descent theorem is Lemma 80.11.3(2), Tag 0ADV: descent data for algebraic spaces  $X_i \rightarrow S_i$  of finite type are effective for any fppf covering family  $\{S_i \rightarrow S\}$ . It yields an algebraic space  $X \rightarrow S$  and compatible  $X_{S_i} \simeq X_i$ . Finite presentation implies finite type, so it applies to our families. We use part (2) explicitly; the unrestricted part (1) has a countability qualification in the Stacks Project's size conventions.

It remains to verify that  $X \rightarrow S$  is an allowed object. For algebraic spaces the following properties are fpqc-local on the target, so they are detected on our cover:

- ◊ finite presentation:  
Lemma 74.11.12, Tag 041V;

- ◊ flatness:

Lemma 74.11.13, Tag 041W;

- ◊ properness:

Lemma 74.11.19, Tag 0422.

Thus  $X \rightarrow S$  has these three properties. In particular it is locally of finite type. Lemma 67.34.3(1), Tag 04NS, says that its relative dimension at a point equals that of any base change at a point above it. Every point of  $X$  lifts to some  $X_{S_i}$  because the base-changed cover is jointly surjective. The local bound at most one therefore holds globally. Preservation under base change alone would not supply this converse; joint surjectivity is also used.

The recovery maps are those of the descent theorem, and two realizations are uniquely isomorphic compatibly by morphism descent. This proves the two missing stack axioms for Curves using explicit geometric inputs. It is the argument packaged in Lemma 99.13.3, Tag 0D1G, and Lemma 99.15.3, Tag 0D52; these packaged results are checks on our argument, not assumptions that Curves is already a stack.

### 19.5 Why algebraic spaces occur

Schemes have fully faithful fpqc descent, but arbitrary scheme descent data need not be effective as schemes. Lemma 110.68.1, Tag 08KF, gives even projective schemes with *étale* descent data that fail to descend in schemes. This is a general example, not specifically a curve-

family example. For curves themselves, Lemma 110.69.1, Tag 0D5E, gives a family over  $\mathbb{A}_k^1$  with total space not a scheme. Allowing algebraic spaces is therefore substantive.

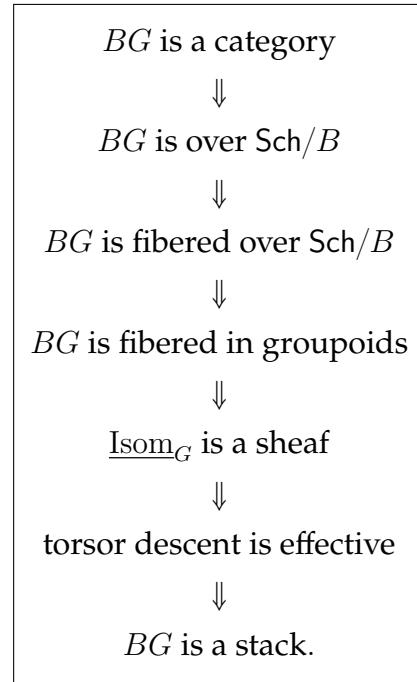
For  $BG$ , affineness made both the local objects and their descended object schemes. For curves, effective descent uses the larger category of algebraic spaces. The Isom axiom only glues arrows between existing objects; the object axiom must construct geometry and check it remains in the class under consideration. This explains why the latter is usually the harder assertion.

**Exercise 22** (Transfer the proof to curves). For each step from category to stack, identify the corresponding step for Curves. Name the extra geometric inputs. Detect the errors in: “Every commutative square of curve families is an arrow” and “Morphisms of schemes descend, so local families always glue to a scheme.”

## 20 Summary of the logical progression

The following chain records the cumulative construction; each downward step includes a new verification, rather than claiming an implication from one isolated

property to the next.



Here is the inventory of black boxes, separated from what the proofs constructed explicitly.

- (1) Fiber products of schemes and their universal property; base change and composition of fppf covers, Tag 0210; and the inclusion of fppf covers among fpqc covers, Tag 022C.
- (2) Subcanonicity and full faithfulness of scheme descent: Tag 023Q, Tag 02W0, and Tag 040L.
- (3) Locality of isomorphisms, affineness, flatness, and local finite presentation: Tag 02L4, Tag 02L5, Tag 02L2, Tag 02KY.

- (4) Effective descent of affine morphisms, Tag 0245. This is the substantial existence theorem for the torsor proof. The action and its equations were then descended explicitly.
- (5) For the interpretation of notation, the quotient-stack definition and torsor comparison, Tag 044Q and Tag 0CQJ. These are not used to prove torsor descent.
- (6) For the curve comparison, morphism descent Tag 04UA, effective descent Tag 0ADV(2), and locality of the defining properties Tag 041V, Tag 041W, Tag 0422, Tag 04NS. Their base-change stability uses Tag 03M0, Tag 04WP, Tag 03XR, and Tag 04NS separately from locality.

The double-cover computations use the elementary finite *étale* algebra criterion (a monic polynomial with invertible derivative) and finite *étale* local splitting. The algebra criterion is Tag 00UB–Tag 00UC; the splitting result is Tag 04HN. Descent of finite locally free morphisms of fixed degree and of *étale* morphisms uses Tag 02V0 and Tag 02VN. Smoothness locality and local sections, used only to compare topologies, are Tag 02VL and Tag 055U. The constant group and  $\mu_2$  were distinguished by their coordinate algebras, not by a characteristic-independent identification.

## 21 Exercises: synthesis and review

Exercises 1–22 appear immediately after the relevant constructions. The next two combine several stages; the final six return to the foundations and can be attempted on a first reading. All exercises have a numbered solution in Section 22.

**Exercise 23** (Change of trivializations). Let  $\tau_i : P_{S_i} \simeq G_{S_i}$  be left-equivariant trivializations, and write  $\tau_j \tau_i^{-1} = R_{h_{ij}}$ . Prove

$$h_{ik} = h_{ij} h_{jk}$$

on triple overlaps. If  $\tau'_i = R_{b_i} \tau_i$ , compute  $h'_{ij}$  and prove the new transitions satisfy the same cocycle. Interpret the answer for  $G = \mathbb{Z}/2\mathbb{Z}$ .

**Exercise 24** (Two failures of forgetting data). Take  $B = \operatorname{Spec} \mathbb{R}$  and  $G = (\mathbb{Z}/2\mathbb{Z})_B$ . Compare the split torsor with  $\operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{R}$ , where the involution is complex conjugation. Prove that they become isomorphic after the fppf cover  $\operatorname{Spec} \mathbb{C} \rightarrow \operatorname{Spec} \mathbb{R}$  but are not isomorphic over  $\mathbb{R}$ . Deduce:

- (a) the prestack of only trivial torsors can satisfy the Isom-sheaf condition while failing effective object descent;
- (b) taking isomorphism classes of all torsors gives a presheaf that need not be a sheaf, although the torsor category is a stack.

### 21.1 Foundational review

**Exercise 25** (Disjoint union versus product). Take  $B = \operatorname{Spec} k$  and  $S = \mathbb{A}_k^1$ . Give the coordinate rings of  $S \amalg S$ ,  $S \times_B S$ , and  $S \times_S S$ . Describe the ring map defining the fold  $S \amalg S \rightarrow S$ . Explain why the fold is an arrow of  $\operatorname{Sch}/B$  and of  $\operatorname{Sch}/S$ , and identify its two sections.

**Exercise 26** (A map into a fiber product). Take  $X = T = \mathbb{A}_k^1$  over  $S = \operatorname{Spec} k$ , and let  $W = \operatorname{Spec} k$  map to the origin of each. Draw the induced map  $W \rightarrow T \times_S X$  and explain why it is not an isomorphism. What extra property of  $W$  would make the comparison an isomorphism? Separately, pull the family  $x^2 = t$  back by  $t = 1$  and compute the result when  $\operatorname{char} k \neq 2$ .

**Exercise 27** (Test points and Yoneda in coordinates). Let  $T = \operatorname{Spec} k[\epsilon]/(\epsilon^2)$ . Show that sending the coordinate of  $\mathbb{A}_k^1$  to 0 and to  $\epsilon$  gives different  $T$ -points with the same image on topological points. For every  $k$ -algebra  $R$ , the rule  $r \mapsto r^2$  is a function on the  $R$ -points of  $\mathbb{A}_k^1$ . Check naturality and recover the scheme morphism by evaluating on the identity point, as in the Yoneda proof.

**Exercise 28** (Three meanings of a group symbol). For  $H = \mathbb{Z}/2\mathbb{Z}$  and  $B = \mathbb{A}_k^1$ , distinguish  $H$ ,  $H_B$ , and  $H_B(T)$ . Describe the unit, multiplication, and inversion of  $H_B$  on its labeled components. Compute  $H_B(T)$  for  $T = B$  and for  $T = B \amalg B$ , and describe the pullback of  $H_B$  to any  $S \rightarrow B$ .

**Exercise 29** (Finite presentation is a relative algebra condition). Explain why  $A[z]/(z^2 - z)$  is a finitely presented  $A$ -algebra and flat as an  $A$ -module. Why is  $A[x]$  finitely presented as an algebra even though, for  $A \neq 0$ , it is not finitely generated as an  $A$ -module? Finally show that  $\mathbb{Z}/2\mathbb{Z}$  is finitely presented as a  $\mathbb{Z}$ -algebra but is not flat over  $\mathbb{Z}$ .

**Exercise 30** (The covering rule in use). Given an fppf cover  $\{S_i \rightarrow S\}$ , a map  $T \rightarrow S$ , and covers  $\{U_{ij} \rightarrow S_i\}_j$ , write the pulled-back and composite covering families. Identify the relevant parts of Tag 0210 and explain why Tag 021N is not the reference for these assertions. For the singleton fold cover  $S \amalg S \rightarrow S$ , compute its pullback to  $T$ , and then cover each of its two sheets by two copies again. Does every arrow in  $(\operatorname{Sch}/B)_{\text{fppf}}$  have to be flat?

## 22 Hints and solutions

The exercise statements occur earlier so that they can be attempted independently. Each solution begins with a short hint and then gives the complete argument.

### Solution to Exercise 1

[[ Hint: First select the objects; then decide which arrows to retain. ]] Both categories have the object collection  $\{x : p(x) = S\}$ . For two objects in this collection, fullness means keeping every arrow:

$$\mathrm{Hom}_{\mathcal{D}}(x, y) = \mathrm{Hom}_{\mathcal{F}}(x, y).$$

The fiber instead keeps precisely the subset on which  $p(u) = \mathrm{id}_S$ . Since the source and target of  $p(u)$  are both  $S$ , a discarded arrow can project to a nonidentity endomorphism of  $S$ .

When  $p = \mathrm{id}_{\mathcal{C}}$ , the equation  $p(x) = S$  forces  $x = S$ . Thus there is one object in each category. Their arrow sets are

$$\begin{aligned}\mathrm{End}_{\mathcal{D}}(S) &= \mathrm{End}_{\mathcal{C}}(S), \\ \mathrm{End}_{\mathcal{F}_S}(S) &= \{\mathrm{id}_S\}.\end{aligned}$$

In particular  $\varepsilon$  is present in  $\mathcal{D}$  and absent from the fiber: the identity *functor* leaves  $\varepsilon$  unchanged; it does not turn it into the identity *morphism*. For  $S = \mathbb{A}_k^1$ , the maps corresponding to  $t \mapsto t^2$  and  $t \mapsto t$  differ as ring maps, hence as scheme maps. This remains true over a finite field even if those maps happen to agree on some set of rational points.

There is no contradiction and no extra triangle condition for an arbitrary functor.

Both categories are subcategories with the same objects, but only the first is required to be full. The word “over  $S$ ” for an object is weaker than the condition “over  $\mathrm{id}_S$ ” for an arrow.

### Solution to Exercise 2

[[ Hint: Both sides of the proposed identity are empty. ]] Both  $G_S \times_S \emptyset$  and  $\emptyset \times_S \emptyset$  equal  $\emptyset$ , so the map between them is an isomorphism. But the empty morphism is not surjective over nonempty  $S$ . No fppf cover can turn it into  $G$ , which has a unit section. The missing requirement is local nonemptiness, supplied here by local triviality or by the surjectivity of an fppf covering.

### Solution to Exercise 3

[[ Hint: Evaluate  $\Theta_P^{-1}$  on  $(p, s(\pi_P(p)))$ . ]] If  $s$  is a section, the map  $g \mapsto g \cdot s$  is equivariant. The torsor identity gives, for every test point  $p$ , a unique  $g$  with  $p = g \cdot s$ , naturally in the test scheme, and hence gives a morphism inverse. A trivial torsor has its unit section. If  $s' = h \cdot s$  for  $h \in G(S)$ , the two trivializations  $F_s(g) = g \cdot s$  obey  $F_{s'} = F_s \circ R_h$ . Every other section has this form, by the same identity.

### Solution to Exercise 4

[[ Hint: Compute idempotents and nilpotents. ]] The constant group has algebra  $k \times k$  and two  $k$ -points. In characteristic 2,  $\mu_2$  has algebra  $k[\epsilon]/(\epsilon^2)$  and only one

$k$ -point: any field element whose square is zero is zero. It also has a nonzero nilpotent element, so its scheme structure differs from the reduced algebra  $k \times k$ . Both algebras have dimension two as  $k$ -vector spaces, but rank does not determine an algebra or a group scheme.

### Solution to Exercise 5

[[ Hint: Apply the composites to a variable  $x$ . ]] We have  $(R_k \circ R_h)(x) = (xh)k = x(hk)$ . Usual groupoid composition sends  $(g_1, g_2)$  to  $g_2g_1$ , but  $(R_{g_2} \circ R_{g_1})(x) = xg_1g_2$ . This reverses the required order. Replacing each parameter by its inverse gives  $g_1^{-1}g_2^{-1} = (g_2g_1)^{-1}$ , exactly the required functorial law.

### Solution to Exercise 6

[[ Hint: Compute the two components separately. ]] For  $a = (f, \tilde{f})$ ,  $b = (g, \tilde{g})$ ,  $c = (h, \tilde{h})$ , both  $(cb)a$  and  $c(ba)$  equal  $(hgf, \tilde{h}\tilde{g}\tilde{f})$ . Both left and right composition with the appropriate identity leave each component fixed. The composites remain equivariant by the calculation in Section 9. Because the top components are actual scheme maps, this is strict associativity of maps. If arrows were specified instead by  $P \simeq f^*Q$ , composition would require the comparison  $f^*g^*R \simeq (gf)^*R$ .

### Solution to Exercise 7

[[ Hint: Evaluate at the unit section of  $G_S$ . ]] The composite of  $e_S : S \rightarrow G_S$  with

the given arrow is a map  $S \rightarrow G_T$  over  $f$ , hence a section  $h \in G(S)$ . Equivariance forces the formula in the exercise, and that formula defines an equivariant morphism for every  $h$ . Under the second arrow,  $xh(s)$  is multiplied on the right by  $k(f(s))$ . Thus the composite over  $gf$  has parameter  $h \cdot f^*k \in G(S)$ , in this order.

### Solution to Exercise 8

[[ Hint: Use the unit parameter in Exercise 7. ]] Take  $\tilde{f} = \alpha_f : G_S \rightarrow G_S$ . The square commutes, and  $\alpha_f$  intertwines the two regular actions over  $f$ . The pair  $(f, \alpha_f)$  is therefore a total arrow. For a nonempty base the endomorphism  $t \mapsto t^2$  of  $\mathbb{A}_B^1$  differs from the identity, so this arrow is not vertical and is excluded from the fiber. If  $B$  is empty the example is vacuous; choose a nonempty base as intended.

### Solution to Exercise 9

[[ Hint: Factor  $u^2 - v^2$  and use that  $2v$  is invertible. ]] The coordinate algebra is

$$k[v^{\pm 1}, u]/(u^2 - v^2).$$

Here  $u$  is automatically invertible. The ideals  $(u - v)$  and  $(u + v)$  are comaximal, so the Chinese remainder theorem identifies the algebra with  $k[v^{\pm 1}] \times k[v^{\pm 1}]$ . The two sections are  $u = v$  and  $u = -v$ . The involution  $u \mapsto -u$  exchanges these sections, giving the trivial double cover of  $T$ .

**Solution to Exercise 10**

[[ Hint: There is no freedom once the two projections are prescribed. ]] The projections must be  $\mathrm{pr}_P \lambda = \tilde{h}$  and  $\mathrm{pr}_T \lambda = k\pi_R$ . They are compatible because  $\pi_P \tilde{h} = f k \pi_R$ . The fiber product gives a unique  $\lambda$ . Applying it to  $g \cdot r$  changes the first component by  $\alpha_h(g)$  and leaves the second unchanged. The induced action on  $P \times_S T$  uses  $\alpha_f \alpha_k(g)$ , which is the same element. Thus the factorization is equivariant. Another arrow over  $k$  satisfying the factorization equation has both the same projections, hence is equal.

**Solution to Exercise 11**

[[ Hint: A pullback includes its arrow to the original object. ]] For a cartesian lift  $u : x \rightarrow y$ , uniqueness concerns vertical automorphisms  $a$  with  $ua = u$ . The cartesian property forces such  $a$  to be  $\mathrm{id}_x$ . For a pulled-back double cover the deck involution changes the projected point of the original cover, so generally  $ua \neq u$ . It is a torsor automorphism but not an automorphism of the prescribed cartesian lift. The latter includes the structure arrow  $u$ .

**Solution to Exercise 12**

[[ Hint: Common refinements trivialize both source and target. ]] If  $U_i$  trivializes  $P$  and  $V_j$  trivializes  $Q$ , then  $U_i \times_S V_j$  jointly form an fppf cover of  $S$ . On this cover the map is  $R_h$  for its value at the unit, hence has inverse  $R_{h^{-1}}$ . Isomorphism

locality proves global invertibility and equivariance of the inverse follows from  $u(gp) = gu(p)$ . Geometric points alone do not detect scheme isomorphisms: for example,  $\mathrm{Spec} k \rightarrow \mathrm{Spec} k[\epsilon]/(\epsilon^2)$  is a bijection on geometric points but not an isomorphism. Finally, the pullback of a trivial torsor along a proper nonempty open immersion of bases gives a total arrow whose top map is an open immersion, not an isomorphism.

**Solution to Exercise 13**

[[ Hint: Choose different automorphisms on the two singleton opens. ]] Restrictions of  $A$  are group homomorphisms. The associated total category has one object over each open, and pullback along inclusions is given by the designated object and the identity group element; the groupoid construction yields cartesian lifts. Every vertical arrow is invertible. On the two singleton opens choose automorphisms 0 and 1. They agree on the empty intersection since  $A(\emptyset)$  is trivial. But a global section of  $A$  is one element that restricts to the same element on both singleton opens. No section has the prescribed pair  $(0, 1)$ , so the automorphism presheaf is not a sheaf.

**Solution to Exercise 14**

[[ Hint: Expand both restrictions and cancel adjacent comparisons. ]] The compos-

ite restriction is

$$\begin{aligned} \rho_v \rho_u(\varphi) &= c_{Q,v}^{-1} v^*(c_{Q,u}^{-1}) \\ &\quad \circ v^* u^* \varphi \circ v^* c_{P,u} \circ c_{P,v}. \end{aligned}$$

Coherence identifies the outer composites with the comparisons for  $uv$ , and identifies the middle map with  $(uv)^* \varphi$ . Thus the result equals  $c_{Q,uv}^{-1} (uv)^* \varphi c_{P,uv}$ . Both sides are maps between the same chosen  $P_{T''}$  and  $Q_{T''}$ , so this is equality, not an unspecified isomorphism of arrows.

### Solution to Exercise 15

[[Hint: Equivariance and inverse laws are equalities of morphisms.]] The maps are  $\varphi a_P$  and  $a_Q(\text{id}_G \times \varphi)$  from  $G_S \times_S P$  to  $Q$ . The theorem's existence clause glues the underlying  $\varphi_i$ , and its uniqueness clause makes these two maps equal since they are equal after the cover. Apply existence again to the compatible  $\varphi_i^{-1}$ , obtaining  $\chi$ . Apply uniqueness to  $\chi\varphi$  and  $\text{id}_P$ , then to  $\varphi\chi$  and  $\text{id}_Q$ . This proves invertibility. A second global isomorphism with the same restrictions has the same underlying map by uniqueness.

### Solution to Exercise 16

[[Hint: All three maps in the cocycle must be over one triple overlap.]] The datum is  $P_i$  and  $\varphi_{ij} : r_1^* P_i \simeq r_2^* P_j$ , with  $q_{23}^* \varphi_{jk} q_{12}^* \varphi_{ij} = q_{13}^* \varphi_{ik}$ . For the constant group the transition is addition of  $\varepsilon_{ij}$ , so composition is addition of the two pulled-back sections. The required equation is

$\varepsilon_{ij} + \varepsilon_{jk} = \varepsilon_{ik}$ . Two swaps give 0, not 1, so the proposed triple gives incompatible identifications whenever that triple overlap is nonempty.

### Solution to Exercise 17

[[Hint: Compare the two tautological sections on the self-overlap.]] Use coordinates on  $P_U$  given by the map  $(x, u) \mapsto x \cdot u$ . On  $U \times_S U$ , the coordinate  $x$  in the first trivialization and  $y$  in the second describe the same point exactly when  $x \cdot u = y \cdot v$ . With  $v = \varepsilon(u, v) \cdot u$ , this says  $y = x + \varepsilon(u, v)$  in  $\mathbb{Z}/2\mathbb{Z}$ . The torsor identity makes  $\varepsilon$  a well-defined section. It is 0 when  $v = u$  and 1 when  $v = \sigma(u)$ . Simple transitivity gives the cocycle by composing these comparisons. Thus the transition is identity along the diagonal but is a swap on the other component. A single covering member still has meaningful descent data on its self-overlap.

### Solution to Exercise 18

[[Hint: Substitute  $\varphi_{ij} = \psi_j \psi_i^{-1}$  into equivariance.]] The equation  $\varphi_{ij} a_i = a_j(\text{id} \times \varphi_{ij})$ , after precomposing with  $\text{id} \times \psi_i$  and postcomposing with  $\psi_j^{-1}$ , becomes  $\psi_i^{-1} a_i(\text{id} \times \psi_i) = \psi_j^{-1} a_j(\text{id} \times \psi_j)$ . These are the transported actions, so they glue. Both associativity maps have domain  $G_S \times_S G_S \times_S P$  and codomain  $P$ ; the unit equation has domain  $P$ . Their equality is checked as maps after the fppf cover. Geometric-point checks alone miss

nilpotents, as in Exercise 12, so they are insufficient.

### Solution to Exercise 19

[[Hint: Both local systems have the same transition map.]] On an overlap, the recovery equations yield  $\psi_j = \varphi_{ij}\psi_i$  and  $\psi'_j = \varphi_{ij}\psi'_i$ , with pullbacks understood. Hence

$$(\psi'_j)^{-1}\psi_j = (\psi'_i)^{-1}\varphi_{ij}^{-1}\varphi_{ij}\psi_i = (\psi'_i)^{-1}\psi_i.$$

The Isom sheaf gives a unique  $\lambda : P \simeq P'$  with  $\psi'_i\lambda_{S_i} = \psi_i$ . If the local identification maps are forgotten, this equation is no longer required, and one can compose with any torsor automorphism. A trivial double cover already has a nonidentity example.

### Solution to Exercise 20

[[Hint: Choose independently on the two connected components.]] Each component admits the identity or the swap, so the global group is  $(\mathbb{Z}/2\mathbb{Z})^2$  with four elements. In general the automorphism sheaf of the trivial torsor is the sheaf of locally constant  $\mathbb{Z}/2\mathbb{Z}$ -valued functions. The value on a disjoint union is the product of the values on its components, exactly as the sheaf axiom requires.

### Solution to Exercise 21

[[Hint: Subtract the two roots in the self-product.]] As a  $k[t]$ -algebra,  $k[x] = k[t][x]/(x^2 - x - t)$  is free with basis  $1, x$ .

The derivative is  $-1$ , so the finite flat map is *étale*. Translation by 1 preserves  $x^2 - x$  and has order two. In the self-product the difference  $y - x$  satisfies  $d(d - 1) = 0$ , and  $(d)$  and  $(d - 1)$  are comaximal. These two components give the torsor identity. A section would yield  $r^2 - r = t$  in  $k[t]$ . For positive degree  $n$ , the left side has degree  $2n$ ; for degree zero it is constant. Neither equals  $t$ . Finally,  $-u = u$  in characteristic 2, so negation does not exchange two sheets and cannot supply the nonidentity constant-group action.

### Solution to Exercise 22

[[Hint: Separate universal properties from closure under descent.]] Define objects and cartesian arrows, then check identity and pasting of squares to obtain a category. Forgetting total spaces is a functor. The fiber product supplies cartesian lifts, while base-change stability of the four geometric conditions ensures that these are allowed objects. A fiber arrow is an isomorphism because its square is cartesian over an identity. The morphism-descent theorem for algebraic spaces, Tag 04UA, glues local isomorphisms and their inverses. Effective descent Tag 0ADV(2) constructs the global algebraic space; the locality theorems for flatness, properness, finite presentation, and relative dimension verify its membership in Curves. An arbitrary commutative square need not be cartesian, so the first proposed assertion has

too many arrows. Full faithfulness for schemes does not give essential surjectivity, and Tag 08KF disproves the second assertion. Affine torsors avoid that problem through Tag 0245; curve families require algebraic spaces.

### Solution to Exercise 23

[[ Hint: Track the order of right multiplications. ]] The composite  $R_{h_{jk}} \circ R_{h_{ij}}$  sends  $x$  to  $xh_{ij}h_{jk}$ , hence  $h_{ik} = h_{ij}h_{jk}$ . The new transition is

$$R_{b_j} \circ R_{h_{ij}} \circ R_{b_i}^{-1},$$

which sends  $x$  to  $xb_i^{-1}h_{ij}b_j$ . Therefore

$$h'_{ij} = b_i^{-1}h_{ij}b_j.$$

On a triple overlap the adjacent factors  $b_jb_j^{-1}$  cancel, giving  $h'_{ij}h'_{jk} = b_i^{-1}h_{ik}b_k = h'_{ik}$ . For  $\mathbb{Z}/2\mathbb{Z}$  this becomes  $\varepsilon'_{ij} = b_i + \varepsilon_{ij} + b_j$ ; changing local labels changes the transition functions but not the glued cover.

### Solution to Exercise 24

[[ Hint: Use  $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathbb{C} \times \mathbb{C}$ . ]] The nonsplit cover is finite *étale* of degree two, with conjugation exchanging the two complex embeddings. Base changing it to  $\mathbb{C}$  gives the algebra  $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathbb{C} \times \mathbb{C}$ , so it becomes the trivial torsor, equivariantly. Over  $\mathbb{R}$  it has no section: no  $\mathbb{R}$ -algebra map  $\mathbb{C} \rightarrow \mathbb{R}$  exists. The split cover has sections, so the covers are not isomorphic.

For (a), transport the canonical descent data of the nonsplit torsor through its trivialization over  $\mathbb{C}$ . This gives a datum of

trivial local torsors. Were it effective in the prestack of trivial torsors, compatible uniqueness in the full torsor stack would make the nonsplit torsor isomorphic to a split one, a contradiction. The arrow sheaves of the trivial-torsor prestack still satisfy the sheaf axiom because they are given by the sheaf  $G$ .

For (b), the two distinct global isomorphism classes restrict to the same local class on this cover. Thus the isomorphism-class presheaf is not even separated. The stack keeps the chosen isomorphisms and their cocycles; passing just to classes discards precisely that information.

### Solution to Exercise 25

[[ Hint: Disjoint union corresponds to a product of rings; fiber product of affine schemes corresponds to a tensor product of rings. ]] The three rings, in the stated order, are

$$k[t] \times k[t], \quad k[t_1, t_2], \quad k[t].$$

The last calculation uses  $k[t] \otimes_{k[t]} k[t] \simeq k[t]$ . The fold corresponds to the diagonal homomorphism  $k[t] \rightarrow k[t] \times k[t]$ ,  $f \mapsto (f, f)$ . Its composite with either projection of rings is the identity, so the two component inclusions are sections of the fold. All maps respect the map to  $\text{Spec } k$ , giving the arrow in  $\text{Sch}/B$ . In  $\text{Sch}/S$  its target is  $(S, \text{id}_S)$  and its source is the disjoint union equipped with the fold as structure map.

**Solution to Exercise 26**

[[Hint: Existence of a comparison map is not universality of its source.]] The fiber product is  $\mathbb{A}_k^2$ . The induced map is its origin, corresponding to  $k[t, x] \rightarrow k$  with  $t, x \mapsto 0$ . This ring map is not an isomorphism, so neither is the scheme map. If  $W$ , with its two projections, satisfied the same universal mapping property for every competing scheme, the comparison would be an isomorphism by the two-factorization argument in Section 3. For the second part the pullback algebra is

$$k[x]/(x^2 - 1) \simeq k \times k,$$

because  $(x - 1)$  and  $(x + 1)$  are comaximal when 2 is invertible. Thus the fiber over  $t = 1$  is two copies of  $\text{Spec } k$ .

**Solution to Exercise 27**

[[Hint: A map into the affine line is determined by the image of its coordinate, including nilpotent information.]] The ring maps  $k[x] \rightarrow k[\epsilon]/(\epsilon^2)$  with images 0 and  $\epsilon$  are different, since  $\epsilon \neq 0$ . Both induce the same map on spectra: the unique prime  $(\epsilon)$  contracts to  $(x)$ . Thus topological-point data alone do not determine a morphism. For any  $k$ -algebra map  $\alpha : R \rightarrow R'$ , one has  $\alpha(r^2) = \alpha(r)^2$ , which is the naturality equation. The identity point of  $\mathbb{A}_k^1$  corresponds to  $x \in k[x]$ . Applying the rule gives  $x^2$ , so the recovered morphism is the squaring map, defined by  $k[x] \rightarrow k[x], x \mapsto x^2$ .

**Solution to Exercise 28**

[[Hint: The labels form a group, the copies of  $B$  form a scheme, and sections choose labels locally constantly.]]  $H$  is the two-element abstract group. The scheme  $H_B$  is two copies  $B_0 \amalg B_1$  of the affine line, with the fold map to  $B$ . The unit includes  $B_0$ , multiplication sends the component  $(a, b)$  to  $B_{a+b}$  by the identity on the base, and inversion is the identity on each component. These are morphisms of  $B$ -schemes. Since  $B$  is nonempty and connected,  $H_B(B) \simeq H$ . The two connected components of  $T = B \amalg B$  can choose their labels independently, giving  $H_B(T) \simeq H \times H$ . Finally  $H_B \times_B S \simeq S \amalg S$ , with exactly the same componentwise group operations over  $S$ .

**Solution to Exercise 29**

[[Hint: Count algebra generators separately from module generators; test flatness by tensoring an injection.]] The first algebra has one generator and one relation. Evaluation at 0 and 1 identifies it with  $A \times A$ , which is free of rank two as an  $A$ -module and hence flat. The algebra  $A[x]$  has one generator and no relations; its module basis is the infinite family  $1, x, x^2, \dots$ . Finitely many polynomials have bounded degrees, so their  $A$ -linear span cannot contain all powers of  $x$  when  $A \neq 0$ . The algebra  $\mathbb{Z}/2\mathbb{Z}$  is  $\mathbb{Z}/(2)$ , hence finitely presented with no variables and one relation. Tensor the injection  $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$  with  $\mathbb{Z}/2\mathbb{Z}$ . The resulting map is the zero

endomorphism of a nonzero group and is not injective. Thus finite presentation does not imply flatness.

### Solution to Exercise 30

[[ Hint: Ask what the new target is in each operation. ]] The pulled-back cover of  $T$  is  $\{T \times_S S_i \rightarrow T\}_i$ ; this is part (3) of Tag 0210. The composite cover of  $S$  is  $\{U_{ij} \rightarrow S_i \rightarrow S\}_{i,j}$ ; this is part (2). Tag 021N instead asserts inclusions between classes of covers, so it does not supply these conclusions. The pullback of the fold is  $T \amalg T \rightarrow T$ . Covering each of these two sheets by two copies gives the four-copy fold onto  $T$  after composition. Doing this instead to the original two  $S$ -sheets gives the four-copy fold onto  $S$ . These are still covers, though a four-copy fold over a nonempty base is not a double-cover torsor for the two-element group. Being a cover and being that particular kind of torsor are different conditions. Finally the arrows of the site are all  $B$ -scheme morphisms. For  $B = \operatorname{Spec} \mathbb{Z}$ , the map  $\operatorname{Spec} \mathbb{F}_2 \rightarrow B$  is an arrow but is not flat, as Exercise 29 shows; it is not an fppf covering map.

## 23 References

[1] The Stacks Project Authors, *The Stacks Project*, [stacks.math.columbia.edu](https://stacks.math.columbia.edu). The linked tags in the text are permanent identifiers; statements and

numbering were checked in September 2026. Principal groups of references:

- ◇ Foundations: fiber products Tag 01J0, Tag 01JQ; product-ring spectra Tag 00ED; relative affine space Tag 01M0, Tag 01LX; affine morphism–ring map correspondence Tag 01I1, Tag 01I2; functors of points Tag 0010, Tag 01J5; Yoneda Tag 001P; sites Tag 00VH; affine morphisms Tag 01S5; local finite presentation Tag 01TP; flatness Tag 01U2.
- ◇ Topologies on Schemes, §34.7 and §34.9: Tag 021M, Tag 021N, Tag 0210, Tag 022B, Tag 022C.
- ◇ Group schemes and actions: Tag 022S, Tag 022T, Tag 047F, Tag 022Z.
- ◇ Groupoid Schemes, §39.11, and Examples of Stacks, §95.14–95.15: Tag 049A, Tag 0499, Tag 049B, Tag 049C, Tag 04UR, Tag 0CQJ.
- ◇ Stacks, §8.2–8.5: Tag 026A, Tag 02ZB, Tag 026B, Tag 026C, Tag 02ZI; quotient-stack definition: Tag 044Q.
- ◇ Categories, §4.33: Tag 02XK, Tag 02XM, Tag 02X0.
- ◇ Descent, §35.13, §35.23, §35.35, §35.37: Tag 023Q, Tag 02L4,

- Tag 02L5, Tag 02L2, Tag 02KY,  
Tag 02W0, Tag 040L, Tag 0245.
- ◇ Smooth and finite *étale*  
examples: Tag 02VL, Tag 055U,  
Tag 02V0, Tag 02VN, Tag 04HN,  
Tag 00UB, Tag 00UC.
  - ◇ Algebraic spaces and curves:  
Tag 04UA, Tag 0ADV, Tag 041V,  
Tag 041W, Tag 0422, Tag 04NS,  
Tag 03M0, Tag 04WP, Tag 03XR,  
Tag 06LR, Tag 0D4Z, Tag 0D1G,  
Tag 0D52.
  - ◇ Limits of scheme descent and  
nonscheme curve families:  
Tag 08KF, Tag 0D5E.
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the two stack conditions; Section 6.3  
discusses canonical pullback  
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StacksForEverybody.tex in the  
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- [3] Bjorn Poonen, *Rational Points on  
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Author-hosted text, Section 2.3,  
especially Definitions 2.3.1 and 2.3.3,  
for  $T$ -valued points and the functor  
of points, and Section 2.3.1 for  
rational points.